Autonomous Spot Cryptocurrency Trading Knowledge Base: Systematic Strategies, Regime Classification, Microstructure, and Risk Architecture

01_core_strategies.md

The following eleven core trading strategies are structured for deterministic parsing and vector-embedding retrieval by an autonomous spot cryptocurrency execution agent operating across Bitcoin (BTC), Ethereum (ETH), and high-liquidity large-cap altcoins. Because spot execution prohibits net short exposure, bearish signals are mapped strictly to SELL or CLOSE actions that liquidate inventory into stablecoin or fiat reserves, whereas bullish signals trigger BUY allocations or HOLD states. Empirical evidence indicates that static indicator rules degrade rapidly in non-stationary cryptocurrency markets; consequently, the strategies below emphasize volatility scaling, volume confirmation, and dynamic trailing exits, which ablation studies demonstrate are the primary drivers of out-of-sample risk-adjusted alpha.

[TREND-01] Adaptive 6-Hour Time-Series Trend with Chandelier ATR Trailing Stop

- Type: trend-following
- Best market regime: High-volatility and sustained trending regimes (60-day BTC return > +15\%); fails in choppy bear markets (60-day BTC return < -15\%, where Sharpe drops to -0.31) and low-volatility sideways consolidations.
- Timeframes: 6-hour (H6) primary execution chart; 1-day (D1) macro regime filter.
- Required inputs: \text{EMA}(20), \text{EMA}(50), \text{ATR}(14), \text{SMA}_{\text{volume}}(20), 60-day rolling asset Sharpe ratio, 60-day BTC return.
- Entry rules: (1) \text{Close}_{t} > \text{EMA}(20)_{t} > \text{EMA}(50)_{t} on H6; (2) \text{Volume}_{t} > 1.5 \times \text{SMA}_{\text{volume}}(20)_{t}; (3) Asset 60-day rolling Sharpe ratio > 0.50. Trigger BUY.
- Exit rules: Trigger CLOSE when \text{Close}_{t} < \text{EMA}(50)_{t} or when the dynamic Chandelier trailing stop is breached: \text{Close}_{t} < \max(\text{High}_{t-19 \dots t}) - \alpha \times \text{ATR}(14)_{t}, where \alpha = 2.5 (calibrated within [2.0, 3.5] monthly).
- Stop-loss rule: Hard initial stop at \text{Entry Price} - 2.5 \times \text{ATR}(14)_{t}.
- Position sizing guidance: Risk 1.0\% of total portfolio equity per trade; scale notional size inversely with \text{ATR}(14)_{t} / \text{Close}_{t}.
- Confirmation / filters: Asset must rank in the top 30 by 30-day median daily volume; 60-day BTC return must exceed -15\%.
- Invalidation conditions: Ignore BUY signal if 8-hour perpetual funding rate exceeds +0.05\% or if H6 candle range exceeds 4.0 \times \text{ATR}(14)_{t} (exhaustion spike) [UNVERIFIED].
- Common failure modes: Whipsaw losses during range-bound consolidations and

parameter decay if lookbacks are not recalibrated monthly.
- Evidence quality: Strong. Out-of-sample evaluation across 150+ pairs (2022–2024) documented an annualized Sharpe ratio of 2.41 and -12.7\% maximum drawdown at 8 bps transaction cost (https://arxiv.org/html/2602.11708v1, https://www.researchgate.net/publication/400742265).

[TREND-02] Multi-Horizon Dual Moving Average Crossover with Volume Filter

- Type: trend-following
- Best market regime: Macro bull trends with persistent institutional inflows; fails in mean-reverting sideways markets where moving average crossovers lag price turns.
- Timeframes: 1-day (D1) primary execution; 4-hour (H4) execution refinement.
- Required inputs: \text{SMA}(20), \text{SMA}(50), \text{SMA}(150), \text{ATR}(14), \text{ADX}(14), \text{SMA}_{\text{volume}}(20).
- Entry rules: (1) \text{SMA}(20)_{t} > \text{SMA}(50)_{t} and \text{SMA}(20)_{t-1} \le \text{SMA}(50)_{t-1}; (2) \text{Close}_{t} > \text{SMA}(150)_{t}; (3) \text{ADX}(14)_{t} > 20.0; (4) \text{Volume}_{t} > 1.25 \times \text{SMA}_{\text{volume}}(20)_{t} [UNVERIFIED]. Trigger BUY.
- Exit rules: Trigger CLOSE when \text{SMA}(20)_{t} < \text{SMA}(50)_{[span_37](start_span)[span_37](end_span)t} or if \text{Close}_{t} falls below a trailing stop of \max(\text{Close}_{\text{since entry}}) - 3.0 \times \text{ATR}(14)_{t}.
- Stop-loss rule: Initial stop at \min(\text{Low}_{t-4 \dots t}) - 1.5 \times \text{ATR}(14)_{t} or \text{Entry Price} - 2.0 \times \text{ATR}(14)_{t}, whichever is higher.
- Position sizing guidance: Risk 0.75\% of equity per trade; cap single-asset weight at w_{\max} = 0.40 of total portfolio value.
- Confirmation / filters: Require D1 \text{ADX}(14) rising (\text{ADX}(14)_{t} > \text{ADX}(14)_{t-3}) [UNVERIFIED].
- Invalidation conditions: Ignore signal if price is more than 15\% above \text{SMA}(50)_{t} at the bar of crossover (late crossover after vertical spike) [UNVERIFIED].
- Common failure modes: Severe lag on entry and exit during V-shaped reversals; high cumulative fee drag if applied to sub-daily timeframes without an ADX filter.
- Evidence quality: Moderate. Documented by Gerritsen et al. (2020) and Detzel et al. (2020) as generating 8.76\% p.a. excess return over buy-and-hold in daily crypto data, though performance requires adaptive window optimization in post-2022 regimes (https://ideas.repec.org/a/eee/finlet/v32y2020ics1544612319308852.html, https://ideas.repec.org/a/gam/jmathe/v13y2025i16p2629-d1725893.html).

[MOM-01] Volatility-Scaled Time-Series Momentum (TSMOM)

- Type: momentum
- Best market regime: Sustained directional macro trends with moderate-to-high volatility; fails during abrupt macro regime reversals and flash crashes.
- Timeframes: 1-day (D1) evaluation and rebalancing.
- Required inputs: Past k-day return r_{t}(k) = (P_{t} / P_{t-k}) - 1 for k \in \{14, 30, 60\}, 30-day realized Garman-Klass volatility \sigma_{\text{GK}}(30), \text{RSI}(14),

\text{ATR}(14).
- Entry rules: (1) Composite momentum score M_{t} = 0.50 \cdot r_{t}(14) + 0.30 \cdot r_{t}(30) + 0.20 \cdot r_{t}(60) > +0.05 (+5.0\%); (2) r_{t}(14) > 0; (3) 50 < \text{RSI}(14)_{t} < 75. Trigger BUY.
- Exit rules: Trigger CLOSE immediately when r_{t}(14) < 0 or M_{t} \le 0 (zero allocation to negative momentum under long-only constraint).
- Stop-loss rule: Hard stop at \text{Entry Price} - 2.2 \times \text{ATR}(14)_{t}.
- Position sizing guidance: Target annualized volatility \sigma_{\text{target}} = 0.25 (25\%). Raw weight w_{i,t} = (\sigma_{\text{target}} / \sigma_{\text{GK}, i, t}(30)) \times \max(M_{i,t}, 0), normalized across active assets such that \sum w_{i,t} \le 1.0 and w_{i,t} \le 0.40.
- Confirmation / filters: Daily turnover must exceed \$50\text{M} USD across spot exchanges; impose a minimum rebalancing threshold of \Delta w_{i,t} > 0.05 to prevent fee erosion.
- Invalidation conditions: Block entry if \sigma_{\text{GK}}(30) > 0.90 (90\% annualized volatility), indicating disorderly market stress.
- Common failure modes: Momentum crashes following parabolic blow-off tops where realized volatility lags sudden price collapse.
- Evidence quality: Strong. Time-series momentum is extensively validated in crypto literature (Hubrich 2017; Liu et al. 2022; 2025/2026 TSMOM benchmarks), yielding Sharpe ratios of 1.30\text{–}1.68, though outperformed by Chandelier-stopped variants (https://arxiv.org/html/2508.16378v2, https://arxiv.org/html/2602.11708v1).

[MOM-02] Cross-Sectional Relative Strength Rotation (Large-Cap Top Quantile)

- Type: momentum
- Best market regime: Risk-on bull and altcoin expansion regimes; fails during liquidity crises and BTC dominance spikes when altcoin correlations converge to 1.0.
- Timeframes: 1-day (D1) or 6-hour (H6) cross-sectional ranking with weekly or 3-day rebalancing.
- Required inputs: Universe of top 15 liquid large-cap crypto assets (excluding stablecoins), normalized moving average trend factor \text{CTREND}_{i,t} = \frac{1}{5}\sum_{L \in \{5, 10, 20, 50, 100\}} (\text{EMA}_{i,t}(L) / P_{i,t}), 7-day return r_{i,t}(7), \text{ATR}(14).
- Entry rules: (1) Rank all universe assets by composite score S_{i,t} = 0.60 \cdot \text{Z-score}(\text{CTREND}_{i,t}) + 0.40 \cdot \text{Z-score}(r_{i,t}(7)); (2) Select top 20\% (top 3 assets out of 15, Long-Only Top-Quantile / LOTQ) ONLY IF r_{i,t}(7) > +0.02 and \text{BTCClose}_{t} > \text{BTC_EMA}(50)_{t}. Trigger BUY.
- Exit rules: Trigger CLOSE when an asset drops below the top 40\% rank at rebalancing or if \text{BTCClose}_{t} < \text{BTC_EMA}(50)_{t}.
- Stop-loss rule: Individual asset stop at \text{Entry Price} - 2.5 \times \text{ATR}(14)_{t} on D1.
- Position sizing guidance: Equal-weight or inverse-volatility weight among the selected top-quantile assets, capped at 25\% of total portfolio equity per altcoin.
- Confirmation / filters: Require aggregate perpetual Open Interest on the selected asset to be flat or rising over 7 days (\Delta \text{OI}_{7\text{d}} \ge 0) to confirm institutional participation.

- Invalidation conditions: Halt all altcoin rotation and stay 100\% cash/BTC if BTC 7-day realized volatility spikes above 70\% annualized during a drawdown.
- Common failure modes: High turnover transaction costs and severe left-tail drawdowns in high-beta altcoins during sudden macro de-risking.
- Evidence quality: Strong. Cross-sectional trend and momentum factors (CTREND, CMOM, LOTQ) demonstrate statistically significant alpha not subsumed by the 3-factor crypto CAPM (https://www.cambridge.org/core/services/aop-cambridge-core/content/view/4C1509ACBA33D5DCAF0AC24379148178/S0022109024000747a.pdf, https://arxiv.org/html/2508.02758v1).

[MOM-03] MACD Zero-Lag Expansion and RSI Regime Momentum

- Type: momentum
- Best market regime: Intraday and multi-day volatility expansions following consolidations; fails in low-volatility choppy ranges.
- Timeframes: 1-hour (H1) and 4-hour (H4).
- Required inputs: \text{MACD}(12, 26, 9) line M_{t}, signal line S_{t}, histogram H_{t} = M_{t} - S_{t}, \text{RSI}(14), \text{EMA}(50), \text{ATR}(14), \text{SMA}_{\text{volume}}(20).
- Entry rules: (1) \text{Close}_{t} > \text{EMA}(50)_{t}; (2) H_{t} > 0 and H_{t} > H_{t-1} > H_{t-2} (expanding positive histogram) with M_{t} > 0; (3) 52 \le \text{RSI}(14)_{[span_175](start_span)[span_175](end_span)[span_179](start_span)[span_179](end_span)t} \le 68 (momentum confirmed without overbought exhaustion); (4) \text{Volume}_{t} > 1.5 \times \text{SMA}_{\text{volume}}(20)_{t}. Trigger BUY.
- Exit rules: Trigger CLOSE when \text{RSI}(14)_{t} > 72, or when H_{t} < 0, or after 12 bars if unrealized profit is < 0.5 \times \text{ATR}(14).
- Stop-loss rule: \text{Entry Price} - 1.8 \times \text{ATR}(14)_{t}.
- Position sizing guidance: Risk 0.50\% of equity per trade due to higher H1/H4 signal frequency [UNVERIFIED].
- Confirmation / filters: Higher-timeframe (D1) \text{Close} > \text{EMA}(50).
- Invalidation conditions: Ignore signal if executed during weekend low-liquidity window (Saturday 00:00 UTC to Sunday 20:00 UTC) when Coinbase/Binance bid-ask spreads widen.
- Common failure modes: False histogram expansions on low-liquidity fakeouts and fee erosion if volume filter (1.5 \times \text{SMA}_{\text{volume}}(20)) is omitted.
- Evidence quality: Moderate. Multi-indicator momentum combinations (MACD, RSI, EMA) are validated in SVM and ensemble classifiers (https://arxiv.org/html/1911.11819v2, https://www.fmdbpub.com/uploads/articles/17922433524__FTSFDS-799-2026.pdf), though standalone unoptimized parameter grids underperform without regime filters.

[MR-01] Bollinger Band and RSI Extreme Oversold Reversion

- Type: mean-reversion
- Best market regime: Ranging, sideways, or bull-market pullback regimes; fails catastrophically during structural bear-market liquidation cascades and stablecoin depegs.
- Timeframes: 1-hour (H1) and 4-hour (H4).

- Required inputs: Bollinger Bands \text{BB}(20, 2.0) (\text{Lower}_{t}, \text{Mid}_{t}, \text{Upper}_{t}), \text{RSI}(14), \text{ADX}(14), \text{EMA}(200)_{\text{D1}}, \text{ATR}(14), \text{SMA}_{\text{volume}}(20).
- Entry rules: (1) \text{Low}_{t-1} < \text{BB_Lower}(20, 2.0)_{t-1} and \text{Close}_{t} > \text{BB_Lower}(20, 2.0)_{t} (re-entry inside band); (2) \text{RSI}(14)_{t-1} < 28 and \text{RSI}(14)_{t} \ge 30; (3) \text{Volume}_{t-1} > 1.5 \times \text{SMA}_{\text{volume}}(20)_{t-1} (capitulation climax volume). Trigger BUY.
- Exit rules: Take 50\% profit at \text{BB_Mid}(20)_{t} (\text{SMA}(20)); CLOSE remaining 50\% at \text{BB_Upper}(20, 2.0)_{t} or when \text{RSI}(14)_{t} > 65, or time-stop after 16 bars.
- Stop-loss rule: Hard stop at \min(\text{Low}_{t-2 \dots t}) - 1.5 \times \text{ATR}(14)_{t}, capped at a maximum price distance of 2.0\% from entry.
- Position sizing guidance: Risk 0.50\% of equity per trade; reduce size by 50\% if D1 \text{Close} < \text{EMA}(200)_{\text{D1}} [UNVERIFIED].
- Confirmation / filters: Require H4 \text{ADX}(14)_{t} < 25.0 to confirm non-trending environment.
- Invalidation conditions: Block entry if perpetual funding rate is > +0.03\% (longs still crowded despite dip) or if exchange liquidation feeds show active cascading long liquidations.
- Common failure modes: "Catching a falling knife" during trend breakdowns where Bollinger Bands expand downward for dozens of consecutive bars.
- Evidence quality: Moderate. Pure RSI/Bollinger mean reversion yields a modest Sharpe of 0.48 standalone, but rises to 1.68 when hybridized with regime and trend filters (https://www.alphaxiv.org/abs/2601.19504, https://arxiv.org/html/2607.12455v1).

[MR-02] Short-Term N-Day Minimum Liquidity Sweep Reversal

- Type: mean-reversion
- Best market regime: Macro uptrends experiencing sharp short-term flushes; fails in sustained macro bear markets (e.g., 2022 drawdown).
- Timeframes: 1-day (D1) signal generation with 4-hour (H4) execution confirmation.
- Required inputs: 20-day rolling minimum \text{MIN}(20)_{t-1} = \min(\text{Low}_{t-20 \dots t-1}), \text{EMA}(200)_{\text{D1}}, \text{RSI}(7)_{\text{D1}}, \text{ATR}(14)_{\text{D1}}.
- Entry rules: (1) Macro filter: \text{Close}_{t} > 0.95 \times \text{EMA}(200)_{\text{D1}}; (2) Intraday sweep: \text{Low}_{t} \le \text{MIN}(20)_{t-1} followed by a daily recovery close \text{Close}_{t} > \text{MIN}(20)_{t-1}; (3) \text{R[span_244](start_span)[span_244](end_span)SI}(7)_{\text{D1}} < 30. Trigger BUY.
- Exit rules: Trigger CLOSE when \text{Close}_{t} > \text{SMA}(10)_{\text{D1}}, or after a fixed holding period of 5 daily bars if \text{Close}_{t} < \text{SMA}(10)_{\text{D1}}.
- Stop-loss rule: \text{Low}_{t} - 1.2 \times \text{ATR}(14)_{\text{D1}} [UNVERIFIED].
- Position sizing guidance: Risk 0.75\% of equity per trade; scale position inversely with 20-day Parkinson volatility.
- Confirmation / filters: Require perpetual funding rate on the asset to flip negative (< 0.00\%) during the sweep of the 20-day low, indicating retail short crowding into support.
- Invalidation conditions: Ignore signal if the 20-day low coincides with a protocol hack, regulatory enforcement action, or stablecoin depeg > 0.5\%.
- Common failure modes: Structural support breakdown transitioning into a multi-month bear trend when the macro filter fails.

- Evidence quality: Moderate. Quantpedia (Padyšák & Vojtko 2022) documented that combining 10–50 day minimum short-term reversal with maximum trend-following on BTC produced a 2.06 return/volatility ratio and -37.67\% MDD, though pre-2022 samples carry structural bull bias (https://quantpedia.com/trend-following-and-mean-reversion-in-bitcoin/).

[BRK-01] Donchian Channel Trading Range Breakout with ATR Regime Filter

- Type: breakout
- Best market regime: Transition from low-volatility consolidation to high-volatility trend expansion; fails in wide, choppy ranges where breakouts immediately mean-revert.
- Timeframes: 4-hour (H4) and 1-day (D1).
- Required inputs: Upper Donchian Channel \text{DC_Upper}(20)_{t-1} = \max(\text{High}_{t-20 \dots t-1}), Lower Donchian Channel \text{DC_Lower}(10)_{t-1} = \min(\text{Low}_{t-10 \dots t-1}), \text{ATR}(14), \text{SMA}_{\text{ATR}}(50) = \text{SMA}(\text{ATR}(14), 50), \text{SMA}_{\text{volume}}(20).
- Entry rules: (1) \text{Close}_{t} > \text{DC_Upper}(20)_{t-1}; (2) Volatility expansion filter: \text{ATR}(14)_{t} > 1.05 \times \text{SMA}_{\text{ATR}}(50)_{t}; (3) Volume trap filter: \text{Volume}_{t} > 1.5 \times \text{SMA}_{\text{volume}}(20)_{t}. Trigger BUY.
- Exit rules: Trigger CLOSE when \text{Close}_{t} < \text{DC_Lower}(10)_{t-1} or when a trailing stop of \max(\text{High}_{\text{since entry}}) - 2.5 \times \text{ATR}(14)_{t} is hit.
- Stop-loss rule: Initial stop at \text{Entry Price} - 2.0 \times \text{ATR}(14)_{t}.
- Position sizing guidance: Risk 1.0\% of equity per trade; apply volatility-adjusted sizing N_{\text{units}} = (0[span_252](start_span)[span_252](end_span).01 \times \text{Equity}) / (2.0 \times \text{ATR}(14)_{t}).
- Confirmation / filters: Require D1 60-day return > 0\% and asset market cap > \$2\text{B} USD.
- Invalidation conditions: Ignore breakout if social volume Z-score > 3.0 on Telegram/X without corresponding spot order-book depth (retail pump fakeout).
- Common failure modes: Bull traps at range highs where stop-hunt sweeps trigger breakout entries right before mean reversion.
- Evidence quality: Moderate. Trading Range Breakout (Donchian) shows positive t-statistics in academic crypto studies (https://arxiv.org/pdf/2010.12415) and improved drawdown control when paired with ATR regime filtering (https://www.researchgate.net/publication/403710434), though raw breakouts deteriorated after 2017 as market structure matured.

[VOL-01] Bollinger Bandwidth Compression and Volatility Expansion Squeeze

- Type: volatility
- Best market regime: Post-compression breakout regimes following multi-day low-volatility coils; fails when volatility expands downward after a false upward wick.
- Timeframes: 4-hour (H4) and 6-hour (H6).
- Required inputs: Bollinger Bands \text{BB}(20, 2.0), Bollinger Bandwidth \text{BBW}_{t} = (\text{[span_55](start_span)[span_55](end_span)BB_Upper}_{t} - \text{BB_Lower}_{t}) /

\text{BB_Mid}_{t}, 100-bar rolling percentile of \text{BBW} (\text{BBW_Pct}(100)), Keltner Channels \text{KC}(20, 1.5 \times \text{ATR}(14)), \text{MACD}(12, 26, 9), \text{SMA}_{\text{volume}}(20).
- Entry rules: (1) Squeeze setup: within the prior 5 bars, \text{BBW_Pct}(100) < 15.0 and \text{BB_Upper}_{t-k} < \text{KC_Upper}_{t-k} [UNVERIFIED]; (2) Release trigger: \text{Close}_{t} > \text{BB_Upper}(20, 2.0)_{t} and \text{BBW}_{t} > 1.15 \times \text{BBW}_{t-1}; (3) \text{MACD_Hist}_{t} > 0 and rising; (4) \text{Volume}_{t} > 1.5 \times \text{SMA}_{\text{volume}}(20)_{t}. Trigger BUY.
- Exit rules: Trail stop at \text{Close}_{t} < \text{EMA}(20)_{t} once unrealized profit exceeds 2.0 \times \text{ATR}(14); take 50\% partial profit at \text{Entry Price} + 3.0 \ti[span_176](start_span)[span_176](end_span)[span_180](start_span)[span_180](end_span)mes \text{ATR}(14)_{t}.
- Stop-loss rule: Hard stop at \text{BB_Mid}(20)_{t} (\text{SMA}(20)) or \text{Entry Price} - 1.8 \times \text{ATR}(14)_{t}, whichever is tighter.
- Position sizing guidance: Risk 0.75\% of equity; because ATR is artificially compressed at entry, use the 30-day median \text{ATR}(14) rather than current-bar \text{ATR}(14) to prevent oversized leverage into a volatility expansion [UNVERIFIED].
- Confirmation / filters: Require perpetual Open Interest to increase by > 3.0\% on the breakout bar alongside neutral or slightly negative prior funding (\le +0.01\%).
- Invalidation conditions: Ignore signal if scheduled macro releases (US CPI, FOMC) occur within 4 hours.
- Common failure modes: "Head-fake" moves where the initial band breach reverses across the 20-period SMA before the true directional trend establishes.
- Evidence quality: Moderate. Volatility compression/expansion factors (weighted at [span_281](start_span)[span_281](end_span)[span_284](start_span)[span_284](end_span)w_{V} = 0.20 in multi-factor crypto architectures) capture episodic volatility spikes that characterize post-2022 crypto markets (https://www.fmdbpub.com/uploads/[span_166](start_span)[span_166](end_span)articles/17922433524__FTSFDS-799-2026.pdf, https://www.mdpi.com/2227-7390/14/5/794).

[RNG-01] Volume-Profile Value Area and Horizontal Range Boundary Reversion

- Type: mean-reversion
- Best market regime: Low-to-moderate volatility horizontal ranging markets (\text{ADX}(14) < 20); fails during macro breakouts and high-velocity news shocks.
- Timeframes: 1-hour (H1) execution within a rolling 14-day Volume Profile structure [UNVERIFIED].
- Required inputs: 14-day Volume Profile Point of Control (\text{POC}), Value Area High (\text{VAH}, encompassing 70\% of volume), Value Area Low (\text{VAL}), \text{RSI}(14), \text{AD[span_64](start_span)[span_64](end_span)X}(14), \text{ATR}(14), 5-bar volume velocity (\text{Volume}_{t} / \text{SMA}_{\text{volume}}(20)_{t}).
- Entry rules: (1) Ranging filter: D1 and H4 \text{ADX}(14)_{t} < 20.0; (2) Price tests or dips up to 0.5 \times \text{ATR}(14) below \text{VAL} and closes back above \text{VAL} (\text{Low}_{t} \le \text{VAL} and \text{Close}_{t} > \text{VAL}); (3) \text{RSI}(14)_{t} < 38; (4) Rebound bar exhibits tape/volume confirmation with \text{Volume}_{t} > 1.3 \times \text{SMA}_{\text{volume}}(20)_{t}. Trigger BUY.

- Exit rules: Take 60\% profit at \text{POC}; CLOSE remaining 40\% at \text{VAH} - 0.25 \times \text{ATR}(14)_{t}, or exit immediately if H4 \text{ADX}(14) crosses above 25.0 [UNVERIFIED].
- Stop-loss rule: Tight fixed structural stop at 1.5\% below entry price (empirically validated optimal range is 1.0\%\text{–}2.0\%).
- Position sizing guidance: Risk 0.60\% of equity per trade; fixed 1.5\% stop sets notional allocation at 40\% of equity (capped by portfolio exposure limits at 25\% per asset, reducing effective risk to 0.375\%).
- Confirmation / filters: Require at least 2 prior rejected touches of the \text{VAL} zone within the preceding 14 days [UNVERIFIED].
- Invalidation conditions: Ignore entry if price approaches \text{VAL} for the 4th time within 48 hours (absorption of resting bid liquidity preceding a breakdown) [UNVERIFIED].
- Common failure modes: Structural breakdown through \text{VAL} when a range transitions into a markdown phase.
- Evidence quality: Moderate. A 5-year backtest (2021–2025) on SOL/USDT demonstrated that Volume Profile mean reversion with tape confirmation and 1.0\%\text{–}2.0\% stop-losses achieved Sharpe ratios > 2.0 and profit factors > 2.5, though results are derived from a single perpetual pair (https://www.researchgate.net/publication/408640724).

[SENT-01] Sentiment-Verified Contrarian Capitulation and Momentum Overlay

- Type: sentiment
- Best market regime: Extreme retail fear capitulations inside macro uptrends or confirmed multi-day sentiment momentum; fails on low-cap tokens subject to coordinated bot/Telegram manipulation.
- Timeframes: 4-hour (H4) and 1-day (D1).
- Required inputs: LLM-verified VADER composite sentiment score $S_{\text{LLM}} \in [-1.0, +1.0]$, 30-day Sentiment Volume Z-score Z_{vol}, Bullish-to-Bearish post ratio R_{BB}, $\text{RSI}(14)$, $\text{SMA}(50)_{\text{D1}}$, $\text{ATR}(14)$.
- Entry rules: Mode A (Contrarian Capitulation): (1) $Z_{\text{vol}} > +2.5$ with extreme bearish sentiment $S_{\text{LLM}} < -0.60$ and $R_{\text{BB}} < 0.35$; (2) Price holds above macro support ($\text{Close}_{t} > 0.90 \times \text{SMA}(50)_{\text{D1}}$) with $\text{RSI}(14)_{\text{H4}} < 30$ followed by a bullish H4 close ($\text{Close}_{t} > \text{Open}_{t}$). Mode B (Verified Trend Confirmation): (1) $\text{Close}_{t} > \text{SMA}(50)_{\text{D1}}$ and $50 < \text{RSI}(14)_{\text{D1}} < 65$; (2) $0.25 \le S_{\text{LLM}} \le 0.65$ with $0.5 \le Z_{\text{vol}} \le 2.0$. Trigger BUY.
- Exit rules: Trigger CLOSE when $S_{\text{LLM}} > +0.80$ paired with $Z_{\text{vol}} > +3.0$ (extreme retail euphoria) or when $\text{RSI}(14)_{\text{H4}} > 75$.
- Stop-loss rule: $\text{Entry Price} - 2.0 \times \text{ATR}(14)_{t}$.
- Position sizing guidance: Risk 0.50\% of equity per trade; sentiment signals act primarily as an allocation multiplier (0.5\times to 1.25\times) on technical base weights.
- Confirmation / filters: LLM verification layer must classify > 80\% of sampled social posts as organic non-bot discourse and confirm zero active Telegram pump countdowns.
- Invalidation conditions: Block entry if social spike is linked to an exchange halt, exploit, or regulatory lawsuit.

- Common failure modes: Over-relying on noisy lexicon scores (raw VADER without LLM context verification) or buying too early into a multi-day fundamental collapse.
- Evidence quality: Moderate. Integrating LLM-verified sentiment with RSI/SMA technical signals inside a mean-variance framework outperforms pure technical baselines (https://arxiv.org/html/2508.16378v2), while contrarian predictive power is empirically concentrated at sentiment extremes (https://arxiv.org/html/2602.07018v3).

02_market_regimes.md

Quantitative Regime Classification Architecture Cryptocurrency time series exhibit pronounced non-stationarity, fat-tailed return distributions, and abrupt structural regime shifts. Empirical analysis from 2020 through 2026 demonstrates that no single trading strategy maintains a positive expectancy across all market environments; for instance, systematic trend-following (TREND-01) achieves an annualized Sharpe ratio of 3.42 in bull regimes and 1.87 in sideways regimes, but degrades to -0.31 in bear regimes. Conversely, adaptive risk and volatility-control frameworks achieve Sharpe ratios up to 3.21 during high-volatility dislocations where static parameter models suffer severe drawdowns. Furthermore, structural market maturation and the growth of spot ETFs and tokenized real-world assets (RWAs) have altered baseline volatility profiles: while Kaiko's broad Standard Crypto Index recorded 51.5\% rolling one-year annualized volatility and its Mid Cap Index stood at 59.0\% in mid-2026, Bitcoin's 7-day realized volatility compressed to 26.0\% during quiet months before surging to 70.0\% during acute June market stress and reverting toward 40.0\%. Consequently, volatility expansions in modern crypto markets are predominantly event-driven and short-lived rather than permanent high-volatility regimes.

To prevent strategy-regime mismatch, the autonomous agent must classify the market state prior to evaluating any entry trigger by computing four orthogonal quantitative dimensions on Bitcoin (the macro benchmark) and the target asset at every 6-hour (H6) and daily (D1) bar close. The first dimension is **Macro Directional Drift (R_{60\text{d}} and EMA Structure)**, which is computed using the 60-day rolling BTC return R_{60\text{d}} = (\text{BTC}_{t} / \text{BTC}_{t-60}) - 1 alongside the daily \text{EMA}(50) and \text{EMA}(200) alignment. Following empirical regime partitioning, the market is classified as a Bull Trend Regime when R_{60\text{d}} > +0.15 (+15\%) and \text{Close}_{\text{D1}} > \text{EMA}(50)_{\text{D1}} > \text{EMA}([span_57](start_span)[span_57](end_span)200)_{\text{D1}}; as a Bear Trend Regime when R_{60\text{d}} < -0.15 (-15\%) and \text{Close}_{\text{D1}} < \text{EMA}(50)_{\text{D1}}; and as a Sideways or Consolidation Regime whenever -0.15 \le R_{60\text{d}} \le +0.15.

The second classification dimension evaluates **Trend Strength versus Mean-Reverting Chop (\text{ADX}(14))** on the D1 and H4 timeframes. An \text{ADX}(14) reading of 25.0 or higher that is actively rising (\text{ADX}(14)_{t} > \text{ADX}(14)_{t-3}) confirms a strong directional trend, whereas readings between 20.0 and 25.0 denote a transitional or developing structure, and readings below 20.0 diagnose a range-bound, mean-reverting environment where trend-following entries suffer elevated whipsaw risk [UNVERIFIED].

The third dimension quantifies **Realized Volatility and Bandwidth Compression (\sigma_{\text{ANN}} and \text{BBW_Pct})**. Because traditional close-to-close volatility ignores intraday price extremes, the agent computes 30-day annualized Garman-Klass volatility \sigma_{\text{GK}}(30)—which incorporates open, high, low, and close prices to achieve 7.4\times greater statistical efficiency than close-to-close estimators in non-trending

markets—alongside drift-adjusted Rogers-Satchell volatility \sigma_{\text{RS}}(30) for trending regimes and the 100-bar rolling percentile of Bollinger Bandwidth \text{BBW_Pct}(100). A Volatility Compression (Squeeze) state is active when \sigma_{\text{GK[span_58](start_span)[span_58](end_span)}}(30)_{\text{BTC}} < 30.0\% annualized or \text{BBW_Pct}(100) < 15.0 [UNVERIFIED]; a Normal Volatility regime spans 30.0\% \le \sigma_{\text{GK}}(30)_{\text{BTC}} \le 60.0\% (or up to 65.0\% for large-cap altcoins, reflecting the 59.0\% mid-cap index baseline); and a High Volatility or Stress Dislocation state is triggered whenever \sigma_{\text{RS}}(30)_{\text{BTC}} > 65.0\% or 7-day realized BTC volatility exceeds 65.0\%.

The fourth dimension captures **Cross-Asset Risk Appetite and Altcoin Beta (\text{BTC.D}, \rho_{\text{EQ}}, and \text{OI})** via the 14-day rate of change in Bitcoin Dominance (\Delta \text{BTC.D}_{14\text{d}}), the 30-day rolling correlation between BTC daily returns and the Nasdaq-100 / S&P 500 (\rho_{\text{EQ}, 30\text{d}}), and aggregate perpetual futures funding rates. A Crypto Risk-On (Altcoin Expansion) regime requires R_{60\text{d}} > +0.15, \Delta \text{BTC.D}_{14\text{d}} < 0, and an 8-hour funding rate between +0.005\% and +0.03\% [UNVERIFIED]. Conversely, when R_{60\text{d}} > +0.15 but \Delta \text{BTC.D}_{14\text{d}} > +1.5\%, the market is in a BTC-Led Defensive Uptrend where institutional capital concentrates in spot BTC ETFs while altcoins lag [UNVERIFIED]. Finally, when R_{60\text{d}} < 0, equity correlation \rho_{\text{EQ}, 30\text{d}} > +0.60 during macro equity sell-offs, or perpetual funding plunges below -0.05\% amidst surging liquidations, the market enters a Macro Risk-Off or Liquidation Cascade state.

Regime-to-Strategy Decision Matrix

The autonomous agent must map the active quantitative regime to strategy activation, strategy suppression, and portfolio cash floors using the following deterministic lookup table:

Regime ID & Name	Quantitative Classification Criteria	Favored Strategy IDs (Active)	Avoided Strategy IDs (Blocked)	Max Portfolio Gross Exposure & Sizing Multiplier
REG-01: Bull Trend (Normal Vol)	R_{60\text{d}} > +15\%, \text{Close}_{\text{D1}} > \text{EMA}(50)_{\text{D1}}, \text{ADX}(14) \ge 25, 30\% \le \sigma_{\text{RS}}(30) \le 65\%	TREND-01, TREND-02, MOM-01, MOM-02, MOM-03, SENT-01 (Mode B)	RNG-01, MR-01 (unless H4 pullback with \text{ADX}_{\text{H4}} < 25)	Max Exposure: 85\% of equity; Sizing Multiplier: 1.00\times
REG-02: Bull Trend (High-Vol Breakout)	R_{60\text{d}} > +15\%, \text{ADX}(14) \ge 25, \sigma_{\text{RS}}(30) > 65\% or \text{ATR}(14) > 1.2 \times	TREND-01, BRK-01, MOM-01 (volatility-scaled)	MR-01, MR-02, RNG-01, TREND-02 (too slow)	Max Exposure: 65\% of equity; Sizing Multiplier: 0.75\times (via inverse-ATR scaling)

Regime ID & Name	Quantitative Classification Criteria	Favored Strategy IDs (Active)	Avoided Strategy IDs (Blocked)	Max Portfolio Gross Exposure & Sizing Multiplier
	\text{SMA}_{\text{ATR}}(50)			
REG-03: Bull Pullback / Dip Sweep	R_{60\text{d}} > +5\%, \text{Close}_{\text{D1}} > \text{EMA}(200)_{\text{D1}}, but 7d return < -8\% and \text{RSI}(14)_{\text{H4}} < 30	MR-01, MR-02, SENT-01 (Mode A)	BRK-01, MOM-02, MOM-03	Max Exposure: 50\% of equity; Sizing Multiplier: 0.60\times [UNVERIFIED]
REG-04: Low-Vol Compression (Squeeze)	-15\% \le R_{60\text{d}} \le +15\%, \sigma_{\text{GK}}(30) < 30\%, \text{BBW_Pct}(100) < 15.0, \text{ADX}(14) < 20	VOL-01 (awaiting band release), RNG-01 (until squeeze fires)	TREND-01, TREND-02, MOM-01, MOM-02	Max Exposure: 40\% of equity; Sizing Multiplier: 0.75\times using 30d median ATR
REG-05: Sideways Horizontal Range	-15\% \le R_{60\text{d}} \l[span_387](start_span)[span_387](end_span)[span_389](start_span)[span_389](end_span)e +15\%, \text{ADX}(14)_{\text{D1}} < 20, 30\% \le \sigma_{\text{GK}}(30) \le 55\%, \text{BBW_Pct}(100) \ge 15.0	RNG-01, MR-01, MR-02	TREND-01, TREND-02, BRK-01, MOM-01, MOM-02	Max Exposure: 50\% of equity; Sizing Multiplier: 0.75\times
REG-06: Bear Market / Macro Markdown	R_{60\text{d}} < -15\%, \text{Close}_{\text{D1}} < \text{EMA}(50)_{\text{D1}} < \text{EMA}(200)_{\text{D1}}, \text{ADX}(14) \ge	SENT-01 (Mode A only at extreme capitulation Z_{\text{vol}} > 3.0), MR-01 (BTC/ETH only, 0.5\times size)	TREND-01, TREND-02, MOM-01, MOM-02, MOM-03, BRK-01, VOL-01, RNG-01	Max Exposure: 15\% of equity (85\%+ Cash/USDC); Sizing Multiplier: 0.35\times

Regime ID & Name	Quantitative Classification Criteria	Favored Strategy IDs (Active)	Avoided Strategy IDs (Blocked)	Max Portfolio Gross Exposure & Sizing Multiplier
	20			
REG-07: Acute Crisis / Liquidation Cascade	24h BTC drop > -10\%, 7d realized vol > 70\%, active exchange halt, or stablecoin depeg > 0.5\%	**NONE** (Force HOLD cash or CLOSE vulnerable positions)	**ALL** (TREND-*, MOM-*, MR-*, BRK-*, VOL-*, RNG-*, SENT-*)	Max Exposure: 0\% new entries; 100\% defensive cash preservation

03_risk_management.md

Position Sizing Methodologies

In a spot-only cryptocurrency portfolio, survival and compounding depend on isolating the portfolio from the extreme right-skewed volatility distribution and left-tail drawdown clusters characteristic of digital assets. To achieve this, the execution engine must compute position size as the minimum of three mathematical models: volatility-adjusted fixed fractional sizing, target volatility scaling, and a Quarter-Kelly empirical cap.

Under the primary default **Volatility-Adjusted Fixed Fractional Sizing** model, let E_{t} denote total account equity (marked to market in USD/USDC), f_{risk} denote the maximum fraction of equity risked on a single trade, P_{entry} denote the execution price, and P_{stop} denote the hard initial stop-loss price (typically $P_{\text{entry}} - m \cdot \text{ATR}(14)_{t}$, where $m \in [1.5, 2.5]$ depending on strategy ID). Defining the stop distance as a percentage of entry price as $d_{\text{stop}} = (P_{\text{entry}} - P_{\text{stop}}) / P_{\text{entry}}$, the notional USD position size V_{pos} is calculated as:

$$V_{\text{pos}} = \frac{E_{t} \cdot f_{\text{risk}} \cdot \lambda_{\text{regime}}}{d_{\text{stop}} + 2 \cdot c_{\text{exec}}}$$

where $\lambda_{\text{regime}} \in [0.35, 1.00]$ is the regime multiplier from 02_market_regimes.md and c_{exec} is the one-way all-in execution cost (fee plus expected slippage, e.g., 0.0010 or 10 bps). The conservative default risk fraction is set at $f_{\text{risk}} = 0.0075$ (0.75\% of total equity per trade) for daily and H6 strategies (TREND-01, TREND-02, MOM-01, BRK-01) and $f_{\text{risk}} = 0.0050$ (0.50\% of equity) for higher-frequency H1 and H4 strategies (MOM-03, MR-01, RNG-01, SENT-01), ensuring that even a cluster of five consecutive losses cannot breach the weekly drawdown ceiling [UNVERIFIED].

For continuous momentum and cross-sectional rotation allocations (MOM-01, MOM-02), fixed fractional sizing is augmented by **Target Volatility Scaling**, wherein notional exposure is scaled inversely with 30-day realized Garman-Klass volatility $\sigma_{\text{GK}, i, t}(30)$ so that no asset's annualized volatility contribution exceeds $\sigma_{\text{target}} = 0.25$ (25\% annualized), bounded by the single-asset concentration cap $w_{\max}$:

$$w_{\text{vol}, i, t} = \min\left(w_{\max}, \frac{\sigma_{\text{target}}}{\sigma_{\text{GK}, i, t}(30)}\right)$$

Finally, when evaluating strategy-level capital allocation via the **Fractional Kelly Criterion**, given historical strategy win rate p and average win-to-loss ratio $b = \overline{W} / |\overline{L}|$, the theoretical Kelly fraction is $f^{*} = p - \frac{1-p}{b}$. However, because cryptocurrency returns exhibit extreme excess kurtosis ($\gamma_{4} \ge 6\text{ to }10$) and negative skewness

($\gamma_{3} < 0$) that violate the continuous Gaussian assumptions of Kelly optimization, full Kelly sizing guarantees catastrophic drawdowns during regime shifts. Therefore, whenever Kelly sizing is referenced as an upper bound on portfolio heat, the agent must enforce a **Quarter-Kelly ($0.25 \cdot f^{*}$)** ceiling computed over a rolling 100-trade out-of-sample window, and if $f^{*} \le 0$, strategy allocation must immediately drop to zero [UNVERIFIED].

Stop-Loss Architecture and Trailing Mechanics

Ablation analysis across 150+ cryptocurrency pairs proves that dynamic trailing stop mechanisms contribute more to risk-adjusted performance ($\Delta \text{Sharpe} = +0.73$, $\Delta \text{MDD} = +9.7\text{ percentage points}$) than market-cap filtering or entry indicator selection. Consequently, every BUY decision must simultaneously register a hard initial stop-loss price P_{stop} in the execution state. For trend and breakout strategies (TREND-01, TREND-02, BRK-01), the stop-loss $P_{\text{stop}, t}$ is updated at the close of every bar using a strict upward **Chandelier ATR Trailing Ratchet**:

$$P_{\text{stop}, t} = \max\left(P_{\text{stop}, t-1}, \max_{k \in [0, 19]}(\text{High}_{t-k}) - \alpha \cdot \text{ATR}(14)_{t}\right)$$

where $\alpha = 2.5$ is the default baseline multiplier. The stop price may never be moved downward; the agent must ignore any calculation that lowers $P_{\text{stop}, t}$. Once unrealized profit on any position reaches $+2.0 \times \text{ATR}(14)_{\text{entry}}$, the hard stop $P_{\text{stop}, t}$ must be raised to at least $P_{\text{entry}} \times (1 + 2.5 \cdot c_{\text{exec}})$ to lock in breakeven plus round-trip fees [UNVERIFIED]. Furthermore, to balance protection against low-liquidity stop-hunting wicks on H1/H4 charts with defense against flash crashes, standard technical stops are triggered on bar Close below $P_{\text{stop}, t}$, while an emergency intrabar hard stop order is placed on the exchange book at $P_{\text{entry}} - 3.5 \times \text{ATR}(14)_{\text{entry}}$ (or -4.0% from entry for RNG-01 and MR-01) to execute immediately intrabar during liquidation cascades [UNVERIFIED].

Portfolio Exposure Caps, Cross-Asset Correlation, and Drawdown Circuit Breakers

Empirical studies confirm a strong long-term price correlation across cryptocurrencies and high-frequency volatility spillover effects (documented via BEKK-MGARCH models by Katsiampa et al.). During market stress, pairwise correlations $\rho_{i,j}$ between BTC, ETH, and large-cap altcoins rapidly approach $0.85\text{–}0.95$, meaning a portfolio holding five altcoins at 20% weight each is effectively a single 100% leveraged bet on the crypto market factor (CMKT). While institutional mean-variance frameworks impose an absolute upper weight limit of $w_{\max} = 0.40$ (40%) per asset, conservative multi-asset live execution requires stricter tiering, correlation clustering, and drawdown circuit breakers as specified below.

Risk Parameter / Guardrail	Conservative Default Threshold	Quantitative Mechanism & Action Rule
Single-Asset Weight Cap ($w_{\max}$)	**BTC / ETH**: 35% of equity (0.35) **Large-Cap Altcoins**: 20% of equity (0.20)	Never allocate more than 35% of total equity to BTC or ETH, or 20% to any single altcoin (stricter than the $w_{\max} = 0.40$ academic baseline) [UNVERIFIED].

Risk Parameter / Guardrail	Conservative Default Threshold	Quantitative Mechanism & Action Rule
Total Altcoin Bucket Cap	40\% of total portfolio equity	Sum of all non-BTC, non-ETH spot positions must not exceed 40\% of total equity due to higher mid-cap volatility (59.0\% vs 51.5\% broad index) [UNVERIFIED].
Maximum Concurrent Open Risk (Portfolio Heat)	3.0\% of total portfolio equity	\sum_{i} \frac{V_{\text{pos}, i} \cdot (P_{\text{entry}, i} - P_{\text{stop}, i})}{P_{\text{entry}, i} \cdot E_{t}} \le 0.030. Positions whose stop-loss has been ratcheted above P_{\text{entry}} contribute 0\% to open risk [UNVERIFIED].
Cross-Asset Correlation Penalty (\rho_{i,j, 30\text{d}})	\rho_{i,j, 30\text{d}} > 0.75	If a candidate asset i exhibits a 30-day rolling hourly return correlation > 0.75 with an existing held asset j, treat i and j as a single combined correlation cluster whose joint weight cannot exceed 35\% of equity and joint open risk cannot exceed 1.25\% [UNVERIFIED].
Daily Drawdown Limit	-3.0\% of start-of-day equity	If realized plus unrealized intraday PnL drops below -3.0\%, trigger CLOSE on all losing positions and lock the agent in HOLD (cash) for 24 hours [UNVERIFIED].
Weekly Drawdown Limit	-6.0\% of rolling 7-day peak equity	Reduce base risk per trade f_{\text{risk}} by 50\% across all strategies and restrict trading exclusively to BTC and ETH for the next 72 hours [UNVERIFIED].
Maximum Portfolio Drawdown Circuit Breaker	-12.0\% from high-water mark	Derived from the -12.7\% empirical maximum drawdown of AdaptiveTrend. At -10.0\% drawdown, halve all position sizes; at -12.0\% drawdown, CLOSE all positions into USD/diversified stablecoins and halt autonomous execution

Risk Parameter / Guardrail	Conservative Default Threshold	Quantitative Mechanism & Action Rule
		pending system recalibration.
Consecutive Losing Streak Protocol	N_{\text{loss}} \ge 3 consecutive losing trades	After 3 consecutive losses: multiply f_{\text{risk}} by 0.75\t[span_102](start_span)[span_102](end_span)imes. After 4 consecutive losses: multiply f_{\text{risk}} by 0.50\times and require top-decile signal confidence. After 5 consecutive losses: pause new entries for 24 hours and re-evaluate 02_market_regimes.md classification. Reset multiplier to 1.0\times only after 2 consecutive winning trades [UNVERIFIED].

04_social_sentiment_playbook.md

Predictive Power and Signal Extraction from Social Media

Academic literature demonstrates that cryptocurrency prices and volatility are closely linked to investor attention and multimodal social sentiment across X (formerly Twitter), Reddit, Telegram, Instagram, and Google Trends search volumes. However, raw lexicon-based sentiment scores (such as standalone VADER or dictionary word counts) suffer from severe noise, sarcasm misinterpretation, and ticker ambiguity. Recent quantitative frameworks solve this via a two-stage pipeline: first computing baseline lexicon scores and attention volume metrics across social feeds, and second passing message windows through an LLM verification layer to validate context and extract exact asset-exchange targets, where LLMs achieve 91.0\% joint extraction accuracy compared to the failure of traditional dictionary heuristics. Platform-specific characteristics also govern signal lead-lag behavior: X and Telegram capture high-frequency breaking narratives and coordination, Reddit reflects retail crowd momentum and community coordination (as documented in the Dogecoin and XRP crowd pumps), while surges in Instagram retail hashtags and Google Trends search volume typically reflect late-stage mainstream retail attention that lags initial institutional spot accumulation [UNVERIFIED]. Crucially, empirical research establishes an asymmetric, regime-dependent relationship between social sentiment and future spot returns. In a **Moderate Sentiment Confirmation (Momentum Regime)**—when social volume is moderately elevated (0.5 \le Z_{\text{vol}} \le 2.0) and LLM-verified sentiment is moderately positive (+0.25 \le S_{\text{LLM}} \le +0.65) alongside bullish price structure (\text{Close} > \text{SMA}(50))—sentiment acts as a confirming factor that improves mean-variance portfolio Sharpe ratios over technical indicators alone. Conversely, in **Extreme Sentiment Tail Regimes**, predictive power in retail social data functions strictly as a **contrarian** indicator. Extreme retail euphoria (S_{\text{LLM}} > +0.80 with social volume spikes Z_{\text{vol}} > +3.0 and surging Google Trends/Instagram retail interest) reliably precedes local distribution tops as late retail buyers exhaust purchasing power, whereas extreme retail panic (S_{\text{LLM}} < -0.60, Z_{\text{vol}} > +2.5) marks short-term seller exhaustion when structural

spot support holds.
To standardize ingest across X, Reddit, Telegram, Instagram, and Google Trends, the agent computes three normalized metrics every hour. First, the **Sentiment Volume Z-Score (Z_{\text{vol}, t})** measures abnormal attention relative to a 30-day rolling baseline:

$$Z_{vol,t} = \frac{V_{mentions,t} - \mu_{mentions}(30d)}{\sigma_{mentions}(30d)}$$

where V_{\text{mentions}, t} is the hourly composite count of deduplicated, bot-filtered mentions and scaled search queries for the asset ticker/name across tracked sources. Second, the **Bullish-to-Bearish Ratio (R_{\text{BB}, t})** measures directional crowd skew:
R_{\text{BB}, t} = \frac{N_{\text{bullish}, t} + 1}{N_{\text{bearish}, t} + 1}
where N_{\text{bullish}, t} and N_{\text{bearish}, t} are the counts of posts classified by the LLM verifier as explicitly bullish (s_{k} > +0.20) or bearish (s_{k} < -0.20) [UNVERIFIED]. Third, **Engagement-Weighted Verified Sentiment (S_{\text{LLM}, t} \in [-1, +1])** computes the weighted average of LLM-verified post polarity scores s_{k} \in [-1, +1], weighted by \ln(1 + \text{UniqueAuthorFollowers}_{k}) and discounted by the author's bot-probability score [UNVERIFIED].

Identifying Manipulated or Coordinated Social Content (Pump-and-Dump vs. Bait-and-Trap)

A critical finding of empirical security research is that the majority of highly active accounts tweeting during alleged cryptocurrency pump events are automated bots, and organized manipulation groups on Telegram and Discord routinely coordinate fraudulent volume spikes. Historically, around 100 organized Telegram channels coordinated an average of 2 pump-and-dump operations per day, generating \$6\text{M} per month in artificial volume (and exceeding \$200\text{M} cumulatively across tracked datasets). These groups operate with a strict hierarchy where admins reveal the target coin to VIP/higher-ranked tiers moments before broadcasting the public countdown and announcement message, causing late retail followers to buy into the insiders' liquidation. Furthermore, longitudinal analysis of 3,677 Telegram-coordinated events reveals a major structural evolution: while 94.5\% of conventional centralized-exchange Pump-and-Dump events occurred prior to June 2022, the post-2022 landscape is dominated (71.5\% of all operations) by **Bait-and-Trap** schemes. In Bait-and-Trap operations, 91.1\% of targeted tokens are deployed within 48 hours preceding the announcement, colluding addresses generate wash-traded volume to mimic a conventional pump, and smart-contract honeypot mechanisms prevent outside buyers from selling, allowing organizers to profit in 99.3\% of operations.
To protect the trading agent from acting as exit liquidity for coordinated Telegram groups, X botnets, or Reddit crowd pumps, the agent must enforce four mandatory manipulation and bot-filtering heuristics in real time. First, under the **Contract Age and Universe Gate**, the agent must never execute a trade on any token deployed within the preceding 180 days—which automatically eliminates the 91.1\% of Bait-and-Trap tokens deployed within 48 hours of an operation—or any asset outside the pre-approved high-liquidity large-cap spot universe. Second, the agent must run a **Telegram Countdown and Announcement Signature Filter**: empirical analysis of 283,017 Telegram messages shows that pump announcements are preceded by explicit countdown messages, average approximately 344 characters in length (roughly 100 characters longer than normal non-pump messages, which average 244 characters), and constitute fewer than 1\% of all channel messages. If an 11-message sliding

window on tracked Telegram channels triggers a real-time pump classifier (such as LightGBM/BGE-M3 achieving F_{1} = 0.83 in milliseconds or the La Morgia et al. detector achieving F_{1} = 94.5\% within 25 seconds) and the LLM extractor identifies a target coin, that asset is immediately placed on a **24-hour BUY blacklist**. Third, the agent must apply a **Bot Swarm and Duplicate-Syntax Heuristic** by computing the ratio of total asset mentions to unique authors (V_{\text{mentions}, t} / N_{\text{unique_authors}, t}) and the cosine similarity of post embeddings within a 5-minute window. If V_{\text{mentions}, t} / N_{\text{unique_authors}, t} > 3.0, or if more than 30\% of posts share near-identical promotional phrasing or originate from accounts with default handles and low follower-to-following ratios, the social spike is classified as an artificial bot swarm and S_{\text{LLM}, t} is forced to 0 [UNVERIFIED]. Fourth, the agent must check for **Social-to-Order-Book Divergence**: if Z_{\text{vol}, t} > +3.0 on X, Reddit, or Instagram while 1\% spot order-book bid depth on tier-1 regulated exchanges (Coinbase, Kraken, Binance) remains flat or declines, the move is classified as an unbacked shill campaign and BUY execution is blocked [UNVERIFIED].

Explicit Sentiment Weighting and Override Rules

Price action and market microstructure always supersede social sentiment; in multi-factor quantitative architectures, technical trend (w_{T} = 0.35), momentum (w_{M} = 0.30), and volatility (w_{V} = 0.20) account for 85\% of decision weight, leaving social and macro sentiment a maximum weight of 15\% (w_{S} = 0.15). The autonomous agent must apply the following deterministic rules to modify or block technical signals:

Sentiment State	Quantitative Thresholds	Mandatory Agent Execution Rule
Extreme Retail Euphoria (Blow-Off Risk)	Z_{\text{vol}, t} > +3.0 AND S_{\text{LLM}, t} > +0.75 AND R_{\text{BB}, t} > 4.0	**1. Block New Entries**: Veto all BUY signals from BRK-01, MOM-01, MOM-02, and MOM-03. **2. Reduce Existing Exposure**: If holding an open position with \text{RSI}(14)_{\text{H4}} > 72, reduce position size by 50\% (SELL half) and tighten the Chandelier trailing stop multiplier \alpha from 2.5 to 1.5 [UNVERIFIED].
Moderate Bullish Euphoria Warning	+2.0 < Z_{\text{vol}, t} \le +3.0 AND S_{\text{LLM}, t} > +0.65	Allow TREND-01 and MOM-01 entries ONLY IF 8-hour perpetual funding is \le +0.02\%, but **reduce position size by 50\%** (\lambda_{\text{sent}} = 0.50\times) [UNVERIFIED].
Organic Bullish Confirmation	+0.5 \le Z_{\text{vol}, t} \le +2.0 AND +0.25 \le S_{\text{LLM}, t} \le +0.65	Confirm technical BUY signals across TREND-01, TREND-02, MOM-01, MOM-03, and SENT-01 (Mode B); apply

Sentiment State	Quantitative Thresholds	Mandatory Agent Execution Rule
		standard 1.00\times position sizing.
Neutral / Low Attention	-1.0 \le Z_{\text{vol}, t} < +0.5 AND -0.25 < S_{\text{LLM}, t} < +0.25	Rely 100\% on OHLCV technical and microstructural rules; sentiment multiplier \lambda_{\text{sent}} = 1.00\times.
Extreme Retail Capitulation (Contrarian Buy Zone)	Z_{\text{vol}, t} > +2.5 AND S_{\text{LLM}, t} < -0.60 AND R_{\text{BB}, t} < 0.35 AND Funding < 0.00\%	Enable SENT-01 (Mode A), MR-01, and MR-02 BUY triggers IF AND ONLY IF spot price closes above a higher-timeframe structural support (\text{Close}_{\text{D1}} > 0.90 \ti[span_319](start_span)[span_319](end_span)[span_323](start_span)[span_323](end_span) mes \text{SMA}(50)_{\text{D1}}) and no fundamental insolvency/hack is detected.
Coordinated Manipulation / Shill Alert	PumpSense / ML Telegram score > 0.80 OR Bot Ratio V_{\text{mentions}} / N_{\text{authors}} > 3.0	**Hard Block**: Immediately reject any BUY signal on the targeted asset for 24 hours; if currently holding the asset into an artificial spike, CLOSE 100\% of the position at market.

05_crypto_market_microstructure.md

24/7 Trading Cycles, Intraday Liquidity, and the Weekend Effect

Although spot cryptocurrency exchanges operate continuously 24 hours a day, 365 days a year, liquidity is heavily fragmented across trading sessions and days of the week. Following the March 2023 closure of the 24/7 institutional USD settlement rails—Silvergate Exchange Network (SEN) and Signature Bank's Signet—and the subsequent surge in spot Bitcoin ETF trading volume in 2024–2026, Bitcoin and large-cap crypto liquidity has structurally shifted from uniform 24/7 activity toward traditional Monday–Friday institutional market hours. Kaiko market structure data reveals that the weekend share of Bitcoin trading volume fell to a record low of 13.0\%, accompanied by significantly wider average weekend bid-ask spreads on U.S. exchanges (such as Coinbase) compared to offshore venues (such as Binance), and a 40\% historical drop in U.S. exchange order-book depth following banking and regulatory transitions versus 25\% offshore. Simultaneously, overall cryptocurrency trading volume in 2026 declined by 52.0\% from its 2025 peak as tokenized assets and RWAs absorbed 29.0\% of perpetual DEX volume (growing from \$4.3\text[span_105](start_span)[span_105](end_span){B} at the start of 2025 to nearly \$30\text{B} on-chain by mid-2026).

To account for these microstructural realities, the autonomous agent must enforce

session-aware execution rules. Non-urgent portfolio rebalancing and large spot orders must be prioritized during the **Peak Liquidity Execution Window** corresponding to the U.S.–European trading overlap (**13:00 UTC to 21:00 UTC**, Monday through Friday), when spot ETF creations/redemptions and institutional market makers are active [UNVERIFIED]. Conversely, during the weekend liquidity trough between **Saturday 00:00 UTC and Sunday 22:00 UTC**, the agent must apply a **Weekend Execution Penalty**: block all breakout (BRK-01) and intraday momentum (MOM-03) entries on H1 and H4 timeframes due to elevated false-breakout risk in thin order books (13\% volume share), reduce the maximum position size on any triggered daily trade by 40\%, and route orders exclusively as limit orders rather than market orders [UNVERIFIED].

Perpetual Futures Spillovers: Funding Rates, Open Interest, and Liquidation Cascades

Even though the agent executes strictly in the **spot** market, perpetual futures (perps) are the most heavily traded instrument in cryptocurrency—invented by BitMEX in 2016 and expanding to a record \$120\text{B} in aggregate open interest in February 2026 (with BTC open interest alone fluctuating around \$54\text{B}\text{–}\$56\text{B}) alongside the 2026 launch of regulated onshore U.S. perpetual futures. Because perp traders routinely operate at 10\times to 50\times+ leverage, where a 10\% adverse move wipes out 10\times margin collateral, forced liquidations in derivatives directly drive spot price dislocations via cross-market basis arbitrage. The perp price is tethered to the underlying spot index by the **Funding Rate**, a periodic fee exchanged typically every 8 hours directly between long and short contract holders. Positive funding indicates that perpetuals trade at a premium to spot and longs pay shorts, whereas negative funding indicates shorts pay longs. During acute short squeezes or capitulations, 8-hour funding can swing violently—for example, flipping from +0.0029\% to -0.0121\%, -0.1902\%, -0.6321\%, and -0.7775\% across consecutive settlements so that shorts pay longs 0.78\% in a single 8-hour window.

When aggregate **Open Interest (OI)** rises rapidly (> +5\% in 24h) alongside a fast rise in positive funding (> +0.03\% per 8 hours), long leverage is dangerously crowded. A minor 3\%\text{–}5\% spot dip triggers initial margin liquidations, forcing automated market sell orders that depress perp prices below spot, inducing arbitrageurs to sell physical spot and buy perps, which cascades into deeper spot liquidations. Conversely, when CME institutional basis is flat, funding rates are neutral, and liquidation heatmaps are sparse, spot price strength reflects genuine physical accumulation. The spot agent must cross-reference derivatives state using the following matrix:

Derivatives Microstructure State	Quantitative Perp / OI Readings	Spot Market Implication & Mandatory Agent Action
Spot-Led Healthy Trend	Spot Price \uparrow, Spot Volume \uparrow, Aggregate OI flat or moderate \uparrow (< +3\%/24h), 8h Funding +0.002\% to +0.015\%	Genuine physical spot accumulation without excessive leverage overhang. Full 1.0\times allocation permitted for TREND-*, MOM-*, and BRK-01.
Overleveraged Long Crowding (Cascade Risk)	Spot Price \uparrow or stalling, Aggregate OI \uparrow > +5\%/24h, 8h Funding >	High probability of imminent long liquidation cascade. **Block all new spot BUY entries**;

Derivatives Microstructure State	Quantitative Perp / OI Readings	Spot Market Implication & Mandatory Agent Action
	+0.03\% (or > +0.05\% on altcoins)	tighten trailing stops on open spot holdings to 1.5 \times \text{ATR}(14) [UNVERIFIED].
Active Long Liquidation Flush (Leverage Reset)	Spot Price \downarrow > -5\%, Aggregate OI \downarrow > -8\%/24h, 8h Funding flips negative (< -0.01\%)	Classic leverage reset. Do not chase the initial falling knife; wait for 1-hour liquidation volume to subside and funding to print < -0.01\%, then enable MR-01 / MR-02 spot entries.
Crowded Short Squeeze Setup	Spot Price flat or rising, Aggregate OI \uparrow > +5\%, 8h Funding deeply negative (< -0.03\% to -0.78\%)	Shorts are paying massive carry and are vulnerable to an upward liquidation squeeze. Favor BRK-01 and VOL-01 upward spot breakouts.

Exchange Outages, Stablecoin Depeg Risk, Slippage Modeling, and Event Risk

Centralized exchanges periodically halt withdrawals or experience matching-engine degradation during network fee surges or banking dislocations—as exemplified when Binance halted BTC withdrawals twice over a single weekend due to surging Bitcoin network fees, or when Binance.US suffered a 40\% drop in market depth and traded at an artificial BTC price premium as customers rushed to trade USD into BTC for faster withdrawal settlement. Similarly, the South Korean "Kimchi premium" normally fluctuates between -1\% and +3\% but spiked to +7\% during the May 2022 Terra collapse and +6\% during the November 2022 FTX collapse. Consequently, the agent must compute all indicator inputs from a **multi-source volume-weighted average price (VWAP)** across at least three independent exchanges (Coinbase, Binance, Kraken) rather than a single venue. If the execution exchange's spot price deviates by > 0.50\% from the composite index, or if the venue halts withdrawals or API acknowledgment latency exceeds 2,000\text{ ms}, the agent must immediately halt trading (HOLD) [UNVERIFIED]. Stablecoins exhibit both downside depeg risk during banking crises (such as USDC depegging following the March 2023 Silvergate/Signet closures) and artificial upside spikes during DeFi borrowing squeezes (such as TrueUSD [TUSD] spiking to \$1.20 on Binance amidst heavy DeFi borrowing). The agent must continuously monitor the spot USD exchange rate of its quote stablecoins (USDC, USDT) on direct fiat venues (Coinbase, Kraken). If |\text{Price}_{\text{stablecoin}} - \$1.0000| > \$0.0050 (50\t[span_341](start_span)[span_341](end_span)[span_343](start_span)[span_343](end_span)ext{ bps}), the agent must freeze all automated strategy signals (HOLD), avoid buying into artificial stablecoin-denominated price pumps that reflect quote-currency debasement rather than asset appreciation, and split idle cash reserves 50\% / 50\% between USDC and direct USD fiat or regulated Treasury-backed tokens [UNVERIFIED].

To prevent transaction costs and market impact from eroding strategy alpha—especially given that average trade sizes declined from \$5.7\text{k} to \$2.8\text{k} on Kraken and from \$3.0\text{k} to \$1.2\text{k} on Coinbase as order books fragmented, while empirical trend strategies require keeping one-way execution costs around 8\text{ bps} (0.08\%) or lower to sustain a Sharpe ratio above 2.0—the agent must model one-way execution cost c_{\text{exec}}

prior to every trade as:
$$c_{\text{exec}} = f_{\text{exchange}} + \frac{s_{\text{bid-ask}}}{2} + \eta \cdot$$
[span_74](start_span)[span_74](end_span)$$\sigma_{\text{1h}} \cdot \sqrt{\frac{V_{\text{pos}}}{\text{Depth}_{1\%}}}$$
where f_{exchange} is the exchange taker/maker fee (2\text{–}10\te[span_75](start_span)[span_75](end_span)xt{ bps}), $s_{\text{bid-ask}}$ is the live quoted spread, $\text{Depth}_{1\%}$ is the aggregated USD order-book depth within 1\% of mid-price, and $\eta \approx 0.10$ is the square-root market impact coefficient [UNVERIFIED]. Any trade whose expected round-trip cost $2 \cdot c_{\text{exec}}$ exceeds 20\% of the expected target profit distance ($\text{TP} - P_{\text{entry}}$) must be rejected (HOLD) [UNVERIFIED]. Finally, the agent must incorporate explicit guardrails for **Macro, Halving, ETF, Unlock, and Regulatory Event Risk**. While Bitcoin's sensitivity to routine macro releases fluctuates across cycles, hawkish Federal Reserve surprises and inflation shocks still trigger abrupt sell-offs (such as BTC sliding 9\% in a week on macro and Fed hawkishness); therefore, new BUY entries are blocked from **2 hours before until 1 hour after** scheduled FOMC rate decisions and U.S. CPI releases [UNVERIFIED]. Regarding **Bitcoin Halvings** and institutional proxy flows (such as corporate treasury accumulation by MicroStrategy holding over 150\text{k} BTC ahead of halving cycles or daily spot ETF net flows), the agent must treat the 72-hour window surrounding a Bitcoin block-reward halving as a high-whipsaw event (requiring a $0.50\times$ sizing reduction) while relying on 02_market_regimes.md ($R_{60\text{d}}$) rather than cyclical narratives to trade post-halving drift [UNVERIFIED]. For **Token Unlocks and New Listings**, empirical data shows that newly launched tokens with massive Fully Diluted Valuations (FDV) relative to circulating market cap—such as SUI launching with a \$600\text{M} circulating market cap against an \$11\text{B} FDV backed by \$300\text{M} in venture funding—suffer sustained post-launch downward price drift as supply unlocks, whereas meme token listings (such as PEPE surging $10\times$ in CEX volume upon OKX listing while still doing half the volume of Uniswap) exhibit extreme blow-off risk. Consequently, the agent must block BUY entries on any asset whose circulating-to-FDV ratio is below 0.20 or within **72 hours prior to and 24 hours after** a scheduled cliff unlock exceeding 2.0\% of circulating supply [UNVERIFIED]. Major **Regulatory Rulings** (such as the landmark July 2023 SEC v. Ripple ruling that propelled XRP to 21[span_532](start_span)[span_532](end_span)\% of global crypto trade volume, surpassing BTC and ETH, while causing divergent open interest flows across SOL, ADA, MATIC versus AXS, MANA, ALGO, and SAND) trigger immediate cross-sectional dispersion; the agent must force an immediate regime and cross-sectional rank recalculation (MOM-02) whenever a major regulatory action occurs.

06_decision_checklist.md

Sequential Pre-Trade Execution Checklist

Before emitting any BUY, SELL, HOLD, or CLOSE instruction, the autonomous trading agent must execute the following six-stage verification sequence in strict numerical order. A failure at any gate immediately terminates the evaluation and forces the specified defensive action.

Step	Verification Gate	Deterministic Pass Criteria	Action on Gate Failure
1	**Infrastructure & Quote Integrity Gate**	(a) Multi-exchange price deviation <	If holding open positions and

Step	Verification Gate	Deterministic Pass Criteria	Action on Gate Failure
		0.50\%; (b) No active withdrawal halt on execution venue; (c) Quote stablecoin price within [\$0.9950, \$1.0050]; (d) Portfolio drawdown from peak > -12.0\% and intraday PnL > -3.0\%.	drawdown breaches -12.0\% or -3.0\% daily limit, emit CLOSE. Otherwise emit **HOLD**.
2	**Open Position & Stop-Loss Audit**	If currently holding the asset: (a) Check if \text{Close}_{t} \le P_{\text{stop}, t} or intrabar low breached emergency stop; (b) Check if strategy exit rule (01_core_strategies.md) is met; (c) Update Chandelier trailing stop P_{\text{stop}, t} upward if new 20-bar high achieved.	If stop or exit rule is breached, emit **CLOSE** (or partial SELL if hitting partial TP target). If stop is intact and no exit triggered, emit **HOLD**.
3	**Macro Regime & Strategy Alignment Gate**	Classify current market regime (REG-01 through REG-07 in 02_market_regimes.md) using R_{60\text{d}}, \text{ADX}(14), and \sigma_{\text{GK}}(30). Verify that the candidate Strategy ID is explicitly listed in the **Favored Strategy IDs** column for the active regime.	Reject candidate signal and emit **HOLD**. Never run a mean-reversion buy in REG-07 or a trend breakout in REG-05.
4	**Signal & Microstructure Confirmation Gate**	(a) All numbered entry conditions for [STRATEGY_ID] are satisfied at bar close t; (b) Volume filter (\text{Volume}_{t} > 1.5 \times \text{SMA}_{\text{volume}}(20)_{t}) is met	Reject candidate signal and emit **HOLD**.

Step	Verification Gate	Deterministic Pass Criteria	Action on Gate Failure
		where required; (c) 8-hour perpetual funding rate is \le +0.03\% (not overleveraged long); (d) Not within weekend H1/H4 block window.	
5	**Social Sentiment & Manipulation Gate**	(a) Zero active Telegram pump/countdown flags (PumpSense score < 0.80) and token age > 180\text{ days}; (b) Bot ratio V_{\text{mentions}} / N_{\text{authors}} \le 3.0; (c) Not in Extreme Retail Euphoria (Z_{\text{vol}} > 3.0 and S_{\text{LLM}} > +0.75).	If manipulation or Extreme Euphoria is detected, block BUY and emit **HOLD** (or CLOSE if holding into a detected pump).
6	**Risk, Correlation & Cost Gate**	(a) Compute V_{\text{pos}} via 03_risk_management.md; (b) Post-trade single-asset weight \le 35\% (BTC/ETH) or \le 20\% (Altcoin) and correlation cluster (\rho_{30\text{d}} > 0.75) weight \le 35\%; (c) Total portfolio open heat \le 3.0\%; (d) Round-trip cost 2 \cdot c[span_499](start_span)[span_499](end_span)_{\text{exec}} \le 20\% of target distance; (e) No CPI/FOMC within 2h.	If all gates 1–6 pass, emit **BUY** with exact notional size V_{\text{pos}} and hard stop P_{\text{stop}}. If Gate 6 fails, emit **HOLD**.

Explicit Conditions Under Which the Correct Action is HOLD

Empirical machine learning classifiers in cryptocurrency trading (such as the monthly-retrained support vector machine framework of arXiv:1911.11819) achieve consistent market outperformance specifically by enforcing a conservative decision-margin threshold (\gamma >

1/3) and **refusing to trade during low-volatility or high-uncertainty intervals**. In a month when BTC plunged 35.6\%, selective execution across only 6 high-confidence trade pairs generated +22.7\% net of fees precisely because the algorithm remained flat (HOLD) during ambiguous bars. Accordingly, the autonomous agent must treat HOLD as its default state whenever multi-timeframe signals conflict—such as when an H1 or H4 momentum trigger (MOM-03) signals BUY while the D1 trend filter (\text{EMA}(50)_{\text{D1}}) slopes downward and D1 \text{RSI}(14) is retreating from overbought levels (> 70). Similarly, HOLD is mandatory during low-volatility equilibrium inside REG-04 (Squeeze) or REG-05 (Range) whenever price hovers in the middle 50\% of the range between \text{VAL} + 0.25(\text{VAH}-\text{VAL}) and \text{VAH} - 0.25(\text{VAH}-\text{VAL}), where neither boundary reversion (RNG-01) nor volatility breakout (VOL-01) edges exist. The agent must also enforce a mandatory HOLD cooldown for at least 2 bars on the execution timeframe after being stopped out of the same asset in the same direction to prevent whipsaw re-entry, as well as whenever open portfolio heat is already \ge 2.5\% or correlated cluster exposure is at its 35\% ceiling [UNVERIFIED].

Common LLM Reasoning Errors and Cognitive Guardrails

Autonomous LLM agents processing financial time series alongside unstructured social text are vulnerable to five recurring reasoning failures that must be suppressed via deterministic constraints. The first error is **Chasing Parabolic Pumps (Recency and FOMO Bias)**: an LLM frequently interprets a massive green candle (> 3.5 \times \text{ATR}(14)) accompanied by surging bullish social commentary as "strong momentum," whereas entering after a multi-ATR vertical expansion violates mean-reversion bounds and places the initial stop-loss far from structural support. To prevent this, the agent is prohibited from triggering BUY on any bar whose range exceeds 3.5 \times \text{ATR}(14) or when price is more than 2.5 standard deviations above \text{SMA}(20), unless executing the initial bar of a validated low-volatility squeeze release (VOL-01) [UNVERIFIED].

The second error is **Overweighting a Single High-Engagement Tweet or Headline (Salience Bias)**: a single viral post on X or Telegram announcement lacks statistical validity and is routinely front-run by insiders or automated bots within seconds. The agent must never initiate a trade based on an individual text message; social information may only influence decisions through the aggregated, bot-filtered Z_{\text{vol}, t} and S_{\text{LLM}, t} distributions defined in 04_social_sentiment_playbook.md.

The third error is **Rationalizing Stop-Loss Overrides (The Disposition Effect)**: on-chain empirical analysis of 370,693k Bitcoin sell transactions across the entire blockchain (2013–2021) confirms that cryptocurrency market participants suffer acutely from the behavioral disposition effect—selling winning positions prematurely while holding onto losing positions during drawdowns in hopes of a rebound. An LLM given discretionary latitude will frequently construct a plausible "macro support" narrative to delay closing a losing trade. Therefore, stop-loss execution in Step 2 of the checklist is mathematically deterministic and non-negotiable; narrative reasoning can never override P_{\text{stop}, t}.

The fourth and fifth errors are **Narrative Hallucination over Quantitative Regime** and **Overconfidence in Backtest Sharpe Ratios (Selection Neglect)**. LLMs trained on financial text often overweight thematic narratives (such as "AI token rallies," "RWA growth," or "halving cycles") even when 60-day returns are deeply negative (REG-06), or treat a published backtest Sharpe ratio above 2.0 as a justification for oversized leverage while ignoring non-normal return tails and multiple-testing bias. Qualitative narratives carry 0\% weight in 02_market_regimes.md, and the agent must never exceed the conservative 0.50\%\text{–}1.00\% per-trade risk caps

regardless of perceived signal confluence.

07_glossary_and_parameters.md

To guarantee zero discrepancy between the quantitative code engine and the LLM decision agent, all indicators, volatility estimators, and evaluation metrics are defined below using exact mathematical formulas and standard default parameters. In all annualization formulas, cryptocurrency continuous trading uses **A = 365 days per year** (unlike 252 in traditional equities), or 365 \times 24 = 8,760 hours for hourly series, or 365 \times 4 = 1,460 bars for 6-hour series.

Indicator / Metric	Standard Parameters	Exact Mathematical Formula & Computational Definition
Log Return (r_{t}) & Simple Return (R_{t}(k))	k \in \{1, 7, 14, 30, 60\}	Log return: r_{t} = \ln(C_{t} / C_{t-1}). k-period simple return: R_{t}(k) = (C_{t} / C_{t-k}) - 1.
Simple Moving Average (\text{SMA})	n \in \{10, 20, 50, 100, 150, 200\}	\text{SMA}(C, n)_{t} = \frac{1}{n} \sum_{i=0}^{n-1} C_{t-i}Applied identically to Volume for \text{SMA}_{\text{volume}}(20).
Exponential Moving Average (\text{EMA})	n \in \{10, 12, 20, 26, 50, 100, 200\}	Multiplier \kappa = \frac{2}{n + 1}. Recursive formula: \text{EMA}(C, n)_{t} = \kappa \cdot C_{t} + (1 - \kappa) \cdot \text{EMA}(C, n)_{t-1}
Normalized Cross-Sectional Trend (\text{CTREND})	L \in \{5, 10, 20, 50, 100\}	\text{CTREND}_{i,t} = \frac{1}{5} \sum_{L \in \{5, 10, 20, 50, 100\}} \frac{\text{EMA}_{i,t}(L)}{C_{i,t}} Captures multi-horizon moving average alignment normalized by current price.
Moving Average Convergence Divergence (\text{MACD})	m = 12 (fast), n = 26 (slow), p = 9 (signal)	MACD Line: M_{t} = \text{EMA}(C, 12)_{t} - \text{EMA}(C, 26)_{t}. Signal Line: S_{t} = \text{EMA}(M, 9)_{t}. Histogram: H_{t} = M_{t} - S_{t}.
Relative Strength Index (\text{RSI})	n = 14 (standard), n = 7 (fast), n = 21 (slow)	Let \Delta C_{t} = C_{t} [span_574](start_span)[span_574](end_span)- C_{t-1}, gain U_{t} = \max(\Delta C_{t}, 0), loss D_{t} = \max(-\Delta C_{t}, 0). Using Wilder's smoothing (\alpha = 1/n): \overline{U}_{t} = \frac{1}{n} U_{t} + \frac{n-1}{n} \overline{U}_{t-1},

Indicator / Metric	Standard Parameters	Exact Mathematical Formula & Computational Definition
		\overline{D}_{t} = \frac{1}{n} D_{t} + \frac{n-1}{n} \overline{D}_{t-1}. \text{RSI}(n)_{t} = 100 - \frac{100}{1 + (\overline{U}_{t} / \overline{D}_{t})}Overbought threshold > 70; oversold threshold < 30.
True Range (\text{TR}) & Average True Range (\text{ATR})	n = 14	$$\text{TR}*{t} = \max\left(H{t} - L_{t},
Average Directional Index (\text{ADX})	n = 14	+\text{DM}_{t} = \max(H_{t} - H_{t-1}, 0) if H_{t} - H_{t-1} > L_{t-1} - L_{t} else 0; -\text{DM}_{t} = \max(L_{t-1} - L_{t}, 0) if L_{t-1} - L_{t} > H_{t} - H_{t-1} else 0. Smooth +\text{DM}, -\text{DM}, and \text{TR} via Wilder's 1/14 decay to get +\text{DI}_{14} = 100 \cdot \frac{\text{Smoothed}(+\text{DM})}{\text{ATR}(14)} and -\text{DI}_{14} = 100 \cdot \frac{\text{Smoothed}(-\text{DM})}{\text{ATR}(14)}. $$\text{DX}_{t} = 100 \cdot \frac{
Bollinger Bands (\text{BB}) & Bandwidth (\text{BBW})	n = 20, k = 2.0	Middle Band: \text{BB_Mid}_{t} = \text{SMA}(C, 20)_{t}. Sample standard deviation: s_{20, t} = \sqrt{\frac{1}{19}\sum_{i=0}^{19}(C_{t-i} - \text{BB_Mid}_{t})^{2}}. Upper/Lower: \text{BB_Upper}_{t} = \text{BB_Mid}_{t} + 2.0 \cdot s_{20, [span_158](start_span)[span_158](end_span)t}, \text{BB_Lower}_{t} = \text{BB_Mid}_{t} - 2.0 \cdot s_{20, t}. Bandwidth: \text{BBW}_{t} = \frac{\text{BB_Upper}_{t} -

Indicator / Metric	Standard Parameters	Exact Mathematical Formula & Computational Definition
		\text{BB_Lower}_{t}}{\text{BB_Mid}_{t}}.
Donchian Channels (\text{DC})	n_{\text{entry}} = 20, n_{\text{exit}} = 10	Computed over prior n completed bars (excluding current bar t to prevent lookahead): \text{DC_Upper}(n[span_107](start_span)[span_107](end_span))_{t-1} = \max_{i \in [1, n]} H_{t-i}, \quad \text{DC_Lower}(n)_{t-1} = \min_{i \in [1, n]} L_{t-i}
Keltner Channels (\text{KC})	n = 20, m = 1.5	\text{KC_Mid}_{t} = \text{EMA}(C, 20)_{t}, \text{KC_Upper}_{t} = \text{KC_Mid}_{t} + 1.5 \cdot \tex[span_85](start_span)[span_85](end_span)[span_87](start_span)[span_87](end_span)t{ATR}(14)_{t}, \text{KC_Lower}_{t} = \text{KC_Mid}_{t} [span_365](start_span)[span_365](end_span)[span_367](start_span)[span_367](end_span)- 1.5 \cdot \text{ATR}(14)_{t}.
Parkinson Volatility (\sigma_{\text{P}})	n = 30\text{ days}, A = 365	High-low range estimator (\sim 5\times more efficient than close-to-close; assumes zero drift): \sigma_{\text{P}}(n)_{t} = \sqrt{\frac{365}{n \cdot 4 \ln(2)} \sum_{i=0}^{n-1} \left(\ln\frac{H_{t-i}}{L_{t-i}}\right)^{2}}
Garman-Klass Volatility (\sigma_{\text{GK}})	n = 30\text{ days}, A = 365	OHLC range estimator (\sim 7.4\times more efficient than close-to-close; optimal in range-bound/sideways markets): \sigma_{\text{G[span_623](start_span)[span_623](end_span)K}}(n)_{t} = \sqrt{\frac{365}{n} \sum_{i=0}^{n-1} \left[0.5 \left(\ln\frac{H_{t-i}}{L_{t-i}}\right)^{2} - (2\ln 2 -

Indicator / Metric	Standard Parameters	Exact Mathematical Formula & Computational Definition
		1)\left(\ln\frac{C_{t-i}}{O_{t-i}}\right)^{2}\right]}where (2\ln 2 - 1) \approx 0.3863 (0.39).
Rogers-Satchell Volatility (\sigm[span_185](start_span)[span_185](end_span)[span_187](start_span)[span_187](end_span)[span_189](start_span)[span_189](end_span)a_{\text{RS}})	n = 30\text{ days}, A = 365	Drift-adjusted OHLC estimator (unbiased in strongly trending markets): \sigma_{\text{R[span_625](start_span)[span_625](end_span)S}}(n)_{t} = \sqrt{\frac{365}{n} \sum_{i=0}^{n-1} \left[\ln\left(\frac{H_{t-i}}{C_{t-i}}\right)\ln\left(\frac{H_{t-i}}{O_{t-i}}\right) + \ln\left(\frac{L_{t-i}}{C_{t-i}}\right)\ln\left(\frac{L_{t-i}}{O_{t-i}}\right)\right]}
Annualized Sharpe Ratio (\text{SR}) & Calmar Ratio (\text{CR})	A = 365 (daily) or 365 \times m (m bars/day)	Given per-period portfolio net returns r_{p, t} and risk-free rate r_{f, t}: $$\text{SR} = \frac{\overline{r_{p} - r_{f}}}{\sigma_{p}} \cdot \sqrt{365 \cdot m}, \quad \text{CR} = \frac{\overline{r_{p}}} \cdot 365 \cdot m}{
Ornstein-Uhlenbeck (OU) Residual Process	\theta_{i} > 0, \mu_{i}, \sigma_{i}	Used in statistical arbitrage/mean-reversion diagnostics: d\rho_{i,t} = \theta_{i}(\mu_{i} - \rho_{i,t})dt + \sigma_{i}dW_{t}, where \theta_{i} is the mean-reversion speed of residual \rho_{i,t} = r_{i,t} - \hat{r}_{i,t}.

08_evidence_and_limitations.md

Empirical Assessment of Technical, Sentiment, and LLM-Driven Trading in Crypto

A rigorous synthesis of post-2020 quantitative literature reveals a sharp structural divide between naive retail technical analysis and adaptive, risk-controlled institutional systems. Early cryptocurrency studies covering 2013–2019 (such as Rohrbach et al. 2017, Hubrich 2017, Corbet et al. 2019, and Gerritsen et al. 2020) documented strong outperformance of simple moving averages and trading range breakouts due to pronounced market inefficiencies and linear retail momentum. However, as institutional participation grew and baseline BTC volatility compressed after August 2022, static fixed-parameter indicator rules suffered substantial decay.

In out-of-sample testing across 150+ cryptocurrency pairs from January 2022 through December 2024, a fixed-parameter trend strategy achieved a Sharpe ratio of 1.34 with a -28.6\% maximum drawdown, whereas incorporating monthly parameter recalibration (\Delta \text{SR} = +1.07) and Chandelier dynamic trailing stops (\Delta \text{SR} = +0.73) elevated the annualized Sharpe ratio to 2.41 and cut maximum drawdown to -12.7\%. Similarly, standalone Bollinger Band and RSI mean reversion yielded a modest Sharpe ratio of 0.48, requiring hybridization with regime and trend filters to reach 1.68, while pure GARCH(1,1) short-term 5-minute volatility strategies performed poorly after transaction costs.
In the domain of social media sentiment, empirical evidence shows that unverified lexicon scores provide negligible predictive power during normal market fluctuations due to bot noise and ticker ambiguity. Statistically robust sentiment alpha is concentrated in two specific applications: using an LLM verification layer to filter VADER sentiment scores within a mean-variance portfolio optimizer, and exploiting **contrarian extremes** where retail euphoria or panic reaches 2.5 to 3.0 standard deviations from baseline alongside real-time detection of Telegram pump-and-dump and Bait-and-Trap frauds. Similarly, recent evaluations of multi-agent LLM frameworks and reinforcement learning (RL) ensembles (2024–2026) confirm that structured agent architectures outperform passive market benchmarks and single-prompt LLM baselines—reducing maximum drawdown by up to 4.17\% and improving Sharpe ratios by 0.21+—precisely when labor is separated into specialized technical, sentiment-verification, and quantitative risk-control modules rather than relying on unconstrained single-prompt text generation.

Backtest Overfitting, Selection Bias, and the Deflated Sharpe Ratio (DSR)

As established by Bailey and López de Prado, a historical backtest is not a controlled scientific experiment and guarantees nothing about future performance because random historical draws do not repeat identically. When a researcher or optimization routine sweeps across N parameter combinations—such as testing 5 RSI lookbacks across 5 entry thresholds (N [span_450](start_span)[span_450](end_span)[span_452](start_span)[span_452](end_span)= 25 trials) or 10 windows across 5 thresholds and 3 exit logics (N = 150 trials)—and selects the configuration with the highest in-sample Sharpe ratio \widehat{\text{SR}}, the reported figure combines any genuine edge with the **largest positive sampling error** in the batch. Because cryptocurrency returns exhibit negative skewness (\gamma_{3} < 0) and fat tails / high kurtosis (\gamma_{4} > 3), standard Sharpe ratio inference severely overstates statistical significance.
To audit any strategy before live deployment, the quantitative pipeline must evaluate two mathematical diagnostics developed by Bailey, Borwein, López de Prado, and Zhu. First, given N independent backtest trials with cross-sectional variance of trial Sharpe ratios V[\{\widehat{\text{SR}}_{n}\}], the expected maximum Sharpe ratio achievable purely by luck under the null hypothesis of zero skill (\text{SR}_{0}) is:
\text{SR}_{0} = \sqrt{V[\{\widehat{\text{SR}}_{n}\}]} \left((1 - \gamma) Z^{-1}\left(1 - \frac{1}{N}\right) + \gamma Z^{-1}\left(1 - \frac{1}{N \cdot e}\right)\right)

$where$\gamma \approx 0.5772$istheEuler - MascheroniconstantandZ^{-1}istheinversestandardnormalcumulativedistr

\text{DSR} \equiv Z\left(\frac{(\widehat{\text{SR}} - \text{SR}_{0})\sqrt{T - 1}}{\sqrt{1 - \hat{\gamma}_{3}\widehat{\text{SR}} + \frac{\hat{\gamma}_{4} - 1}{4}\widehat{\text{SR}}^{2}}}\right)
A strategy configuration is statistically validated only if \text{DSR} \ge 0.95. Empirical worked

examples demonstrate that an annualized Sharpe ratio of 1.35 selected from N = 25 trials (T = 1500, \hat{\gamma}_{3} = -0.5, \hat{\gamma}_{4} = 6) or even an annualized Sharpe ratio of 2.50 selected from N = 100 trials (T = 1250, \hat{\gamma}_{3} = -3, \hat{\gamma}_{4} = 10) yields \text{DSR} \approx 0.90, failing the 0.95 significance bar due to multiple testing and fat tails. Second, **Combinatorially Symmetric Cross-Validation (CSCV)** partitions the performance matrix M (T \times N) into S even submatrices (e.g., S = 16, generating \binom{16}{8} =[span_18](start_span)[span_18](end_span)[span_29](start_span)[span_29](end_span) 12,870 train/test splits), records the relative out-of-sample rank \bar{\omega}_{c} \in (0, 1) of the in-sample winner n^{*}, and integrates the distribution of logits \lambda_{c} = \ln\left(\frac{\bar{\omega}_{c}}{1 - \bar{\omega}_{c}}\right) below zero to compute the **Probability of Backtest Overfitting (\text{PBO} = \int_{-\infty}^{0} f(\lambda)d\lambda)**. Any strategy optimization pipeline exhibiting \text{PBO} > 0.20\text{–}0.25 must be rejected as an artifact of noise.

Paper Trading vs. Live Execution Divergences

Live spot cryptocurrency execution systematically underperforms paper backtests due to four structural frictions that the agent must continuously account for. First, **Survivorship Bias in Historical Universe Selection** inflates backtest returns whenever simulations are run solely on tokens that survived to the present day, omitting former top-cap assets (such as Terra's LUNA/UST, FTX's FTT, Celsius's CEL, or Multichain) that suffered -99\% terminal collapses. Second, **Non-Linear Slippage and Queue Position during Volatility Spikes** create execution drag: whereas paper trading assumes frictionless fills at the bar close price C_{t}, live breakout (BRK-01) and stop-loss orders execute precisely when top-of-book liquidity evaporates, particularly during weekends (13\% BTC volume share) and on U.S. exchanges where depth declined 40\%. Third, **Lookahead Bias in Multi-Timeframe and Volume-Profile Features** occurs when daily indicators or rolling volume profiles are evaluated using unclosed higher-timeframe bars, leaking future intraday extremes into hourly signals; the execution code must strictly reference t-1 completed higher-timeframe bars. Fourth, **API Latency and Exchange Matching Delays** degrade stop-loss fills during liquidation cascades or network fee surges, when exchange WebSocket feeds lag and prices gap through theoretical stop levels.

Synthesis of the 10 Foundational Sources and Empirical Contributions

The quantitative rules, thresholds, and architectural constraints across this knowledge base are anchored in the following ten foundational research sources:

Rank	Source Citation & Direct Link	Core Empirical & Methodological Contribution to the Knowledge Base
1	**Systematic Trend-Following with Adaptive Portfolio Construction (AdaptiveTrend, 2026)** https://arxiv.org/html/2602.11708v1 /	Provided the primary H6 trend-following architecture (TREND-01), regime-conditional performance benchmarks (Bull SR 3.42, Sideways SR 1.87, Bear SR

Rank	Source Citation & Direct Link	Core Empirical & Methodological Contribution to the Knowledge Base
	https://www.researchgate.net/publication/400742265	-0.31), 8 bps transaction cost resilience, and the ablation proof that dynamic Chandelier trailing stops (\Delta \text{SR} = +0.73) and monthly recalibration (\Delta \text{SR} = +1.07) drive out-of-sample crypto alpha.
2	**Sentiment-Enhanced Portfolio Optimization with LLM Verification (2025/2026)** https://arxiv.org/html/2508.16378v2	Established the architecture for SENT-01 and MOM-01, proving that verifying VADER sentiment scores via an LLM context layer and combining them with \text{RSI}(14) and \text{SMA} signals under a w_{\max} = 0.40 concentration cap and turnover constraint outperforms standalone technical baselines.
3	**Trend Factor for the Cross Section of Cryptocurrency Returns (JFQA / Cambridge, 2024)** htt[span_379](start_span)[span_379](end_span)[span_383](start_span)[span_383](end_span)ps://www.cambridge.org/core/services/aop-cambridge-core/content/view/4C1509ACBA33D5DCAF0AC24379148178/S0022109024000747a.pdf	Validated cross-sectional trend (\text{CTREND}) and momentum (\text{CMOM}) factors across cryptocurrencies (MOM-02), demonstrating that multi-horizon moving average signals generate statistically significant alpha (3.87\% weekly spread) not explained by the 3-factor crypto CAPM.
4	**CTBench: Cryptocurrency Time Series Benchmark & Canonical Strategies (2025)** https://arxiv.org/html/2508.02758v1	Defined the Long-Only Top-Quantile (LOTQ) cross-sectional allocation rule (MOM-02), Ornstein-Uhlenbeck (OU) mean-reversion residual modeling, and the right-skewed hourly volatility distribution across large-, mid-, and small-cap crypto assets.
5	**Multi-Factor Indicator Spectrum & Micro-Dislocation Triggers (FTSFDS, 2026)** https://www.fmdbpub.com/uplo	Supplied the explicit factor bucket weightings (Trend w_{T} = 0.35, Momentum w_{M} = 0.30, Volatility w_{V} = 0.20), standard indicator lookbacks

Rank	Source Citation & Direct Link	Core Empirical & Methodological Contribution to the Knowledge Base
	ads/articles/17922433524__FTSFDS-799-2026.pdf	(\text{EMA} 10–200, \text{RSI} 7/14/21, \text{ATR}, \text{BBW}, Donchian), and the mandatory volume-trap filter (\text{Volume}_{t} > 1.5 \times \text{SMA}_{\text{volume}}(20)) used across TREND-01, MOM-03, MR-01, BRK-01, and VOL-01.
6	**Volume Profile Mean Reversion with Tape Speed Confirmation on SOL/USDT (2026)** https://www.researchgate.net/publication/408640724	Grounded the horizontal range strategy (RNG-01), providing 5-year empirical backtest evidence that volume-profile mean reversion paired with volume velocity confirmation and tight 1.0\%\text{–}2.0\% stop-losses achieves Sharpe ratios > 2.0 and profit factors > 2.5.
7	**Uncovering Bait-and-Trap Schemes & Real-Time Telegram Pump Detection (2023–2026)** https://arxiv.org/html/2610.10149v1, https://arxiv.org/html/2605.09431v1, https://arxiv.org/html/2105.00733v2	Provided forensic rules for 04_social_sentiment_playbook.md: documented that 71.5\% of post-2022 Telegram schemes are Bait-and-Trap honeypots (91.1\% deployed within 48h prior), while ML detectors (La Morgia et al., 94.5\% F1 in 25s) and LLMs (PumpSense, 91.0\% extraction accuracy) reliably flag manipulation.
8	**Contrarian Predictive Power of Extreme Social Sentiment in Bitcoin (2026)** https://arxiv.org/html/2602.07018v3	Established the empirical basis for using retail social sentiment as a **contrarian** indicator at market extremes, directly informing the position-sizing reduction and entry-blocking rules during extreme retail euphoria in 04_social_sentiment_playbook.md.
9	**Kaiko & CryptoRank Market Microstructure, Volatility, and Liquidity Reports (2023–2026)**	Provided the institutional microstructure benchmarks for 05_crypto_market_microstructure.md: the record-low 13.0\%

Rank	Source Citation & Direct Link	Core Empirical & Methodological Contribution to the Knowledge Base
	https://www.theblock.co/news/markets/2024-02-27-bitcoin-weekend-trading-279353, https://www.kaiko.com/resources/xrp-volume-surpasses-btc-following-landmark-ruling, https://cryptorank.io/insights/research/crypto-cards-are-turning-stablecoins-into-everyday-payments	weekend BTC volume share, wider weekend U.S. spreads, -40\% U.S. vs -25\% offshore depth drops, stablecoin depeg anomalies, and 2026 realized volatility compression (26\% quiet BTC vol vs 51.5\% broad index and 59\% mid-cap index).
10	**The Deflated Sharpe Ratio & Probability of Backtest Overfitting (Bailey & López de Prado, 2014–2025)** https://www.davidhbailey.com/dhbpapers/backtest-prob.pdf, https://saral.money/blog/backtest-overfitting-deflated-sharpe/	Supplied the mathematical formulas and statistical guardrails (\text{DSR} \ge 0.95, \text{PBO} \le 0[span_110](start_span)[span_110](end_span).20 via CSCV) in 08_evidence_and_limitations.md to correct backtest Sharpe ratios for multiple-testing selection bias, small sample lengths, negative skewness, and fat-tailed kurtosis.

Works cited

1. Forward Machine Learning Filtering for Cryptocurrency Markets, https://www.fmdbpub.com/uploads/articles/17922433524__FTSFDS-799-2026.pdf 2. (PDF) Systematic Trend-Following with Adaptive Portfolio Construction, https://www.researchgate.net/publication/400742265_Systematic_Trend-Following_with_Adaptive_Portfolio_Construction_Enhancing_Risk-Adjusted_Alpha_in_Cryptocurrency_Markets 3. Review and Applications of Cryptocurrency Algorithmic Trading, https://cse.aua.am/wp-content/uploads/2025/06/Capstone-final.pdf 4. Sentiment-Aware Mean-Variance Portfolio Optimization for ... - arXiv, https://arxiv.org/html/2508.16378v2 5. Cryptocurrency Price Prediction and Trading Strategies Using, https://arxiv.org/html/1911.11819v2 6. CTBench: Cryptocurrency Time Series Generation Benchmark - arXiv, https://arxiv.org/html/2508.02758v1 7. Systematic Trend-Following with Adaptive Portfolio Construction, https://arxiv.org/html/2602.11708v1 8. Optimizing Crypto-Trading Performance: A Comparative Analysis of, https://www.mdpi.com/2227-7390/14/5/794 9. fundingrates Community Insights & Market Sentiment - Binance, https://www.binance.com/en/square/hashtag/fundingrates 10. TREND FOLLOWING COVEL Full Version - arxiv.andrew-silva.com, http://arxiv.andrew-silva.com/paper/uFU5Y7CN90OH/Trend-Following-Covel 11. Technical trading rules' profitability and dynamic risk premiums of, https://pdfs.semanticscholar.org/7d6a/955a163841b7082af1c832e57abfd8078f1a.pdf 12.

Technical trading rules in the cryptocurrency market - IDEAS/RePEc, https://ideas.repec.org/a/eee/finlet/v32y2020ics1544612319308852.html 13. Evaluating the Performance of a Donchian Channel Breakout, https://www.researchgate.net/publication/403710434_Evaluating_the_Performance_of_a_Donchian_Channel_Breakout_Strategy_with_ATR-Based_Risk_Management 14. Adaptive Optimization of a Dual Moving Average Strategy for, https://ideas.repec.org/a/gam/jmathe/v13y2025i16p2629-d1725893.html 15. Trend-following and Mean-reversion in Bitcoin - QuantPedia, https://quantpedia.com/trend-following-and-mean-reversion-in-bitcoin/ 16. Momentum and trend following trading strategies for currencies and, https://assets.super.so/e46b77e7-ee08-445e-b43f-4ffd88ae0a0e/files/9c27aa78-9b14-4419-a53d-bc56fa9d43b2.pdf 17. Range Volatility Calculator | Parkinson, Garman-Klass & Rogers, https://ryanoconnellfinance.com/calculators/range-volatility-calculator/ 18. CLaC @ FinMMEval 2026 Task 3: Sentiment-Augmented Deep, https://arxiv.org/html/2607.16028v1 19. Crypto Cards Are Turning Stablecoins Into Everyday Payments, https://cryptorank.io/insights/research/crypto-cards-are-turning-stablecoins-into-everyday-payments 20. A Trend Factor for the Cross Section of Cryptocurrency Returns, https://www.cambridge.org/core/services/aop-cambridge-core/content/view/4C1509ACBA33D5DCAF0AC24379148178/S0022109024000747a.pdf/trend_factor_for_the_cross_section_of_cryptocurrency_returns.pdf 21. XRP Volume Surpasses BTC Following Landmark Ruling - Kaiko, https://www.kaiko.com/resources/xrp-volume-surpasses-btc-following-landmark-ruling 22. Volume Profile Mean Reversion Strategy with Tape Speed, https://www.researchgate.net/publication/408640724_Volume_Profile_Mean_Reversion_Strategy_with_Tape_Speed_Confirmation_for_Cryptocurrency_Futures_Markets_A_5-Year_Backtest_Study_on_SOLUSDT_Perpetual_Contracts 23. As spot ETFs grow, bitcoin liquidity transforms from 24/7 ... - The Block, https://www.theblock.co/news/markets/2024-02-27-bitcoin-weekend-trading-279353 24. EVOQUANT: Self-Evolving Verifier-Guided Strategy Optimization for, https://arxiv.org/html/2607.12455v1 25. CSTrader: A Testbed for Language-Grounded Trading in a ... - arXiv, https://arxiv.org/html/2606.31461v1 26. Generating Alpha: A Hybrid AI-Driven Trading System Integrating, https://www.alphaxiv.org/abs/2601.19504 27. How To Compute Volatility 6 Ways Most People Don't Know, https://www.pyquantnews.com/the-pyquant-newsletter/how-to-compute-volatility-6-ways 28. TUSD Depegs Amidst Heavy Borrowing - Kaiko, https://www.kaiko.com/resources/tusd-depegs-amidst-heavy-borrowing 29. Exploring Investor Behavior in Bitcoin: A Study of the Disposition Effect, https://arxiv.org/pdf/2010.12415 30. Identifying and Analyzing Cryptocurrency Manipulations in Social, https://arxiv.org/pdf/1902.03110 31. Bitcoin Slides Towards $27K, Down 9% Over the Week - Decrypt.Co, https://decrypt.co/137662/bitcoin-slides-towards-27k-down-9-over-the-week 32. Detection and Analysis of Telegram Bait-and-Trap Schemes - arXiv, https://arxiv.org/html/2610.10149v1 33. 1 Introduction - arXiv, https://arxiv.org/html/2602.07018v2 34. The Anatomy of a Cryptocurrency Pump-and-Dump Scheme - arXiv, https://arxiv.org/html/1811.10109v2 35. Binance Halts Withdrawals as BTC Fees Surge - Kaiko, https://www.kaiko.com/resources/binance-halts-withdrawals-as-btc-fees-surge 36. AutoQuant: An Auditable Expert-System Framework for Execution, https://arxiv.org/html/2512.22476v3 37. Revisiting Ensemble Methods for Stock Trading and Crypto ... - arXiv, https://arxiv.org/html/2501.10709v1 38. Deflated Sharpe Ratio (DSR). Balaena Quant Insights:

Issue 24, https://medium.com/balaena-quant-insights/deflated-sharpe-ratio-dsr-33412c7dd464 39. Backtest Overfitting: The Deflated Sharpe Test - Saral, https://saral.money/blog/backtest-overfitting-deflated-sharpe/ 40. Backtest Overfitting and Deflated Sharpe Ratio | PDF - Scribd, https://www.scribd.com/document/756967223/ssrn-2606462 41. Cryptocurrency Trading: A Comprehensive Survey - arXiv, https://arxiv.org/html/2003.11352v5 42. Research on the Price Prediction Algorithms of Major ... - arXiv, https://arxiv.org/html/2609.18149 43. The Anatomy of a Cryptocurrency Pump-and-Dump Scheme, https://ideas.repec.org/p/arx/papers/1811.10109.html 44. The dynamics of cryptocurrency market behavior: sentiment analysis, https://www.emerald.com/imds/article/122/2/365/183656/The-dynamics-of-cryptocurrency-market-behavior 45. Real-Time Detection and Target Extraction of Crypto Pump ... - arXiv, https://arxiv.org/html/2605.09431v1 46. The Doge of Wall Street: Analysis and Detection of Pump and Dump, https://arxiv.org/html/2105.00733v2 47. Don't You Know, Pump it Up! Investigating Cryptocurrency ... - arXiv, https://arxiv.org/abs/2609.01176 48. The Aftermath: What Happened to USDC? | Traders' Insight, https://www.interactivebrokers.com/campus/traders-insight/the-aftermath-what-happened-to-usdc/ 49. What are perpetual futures? Perps, funding rates, and liquidations, https://www.bitget.com/news/detail/12560605471776 50. Crypto Derivatives Open Interest Hits Record $120 Billion In, https://www.onebullex.com/news/articles/crypto-derivatives-open-interest-hits-record-120-billion-in-february-2026-3 51. Crypto Perpetuals Funding - ML for Trading, https://ml4trading.io/case-studies/crypto-funding-arbitrage/ 52. THE PROBABILITY OF BACKTEST OVERFITTING - David H Bailey, https://www.davidhbailey.com/dhbpapers/backtest-prob.pdf 53. 8.3 The Dangers of Backtesting - Portfolio Optimization Book, https://portfoliooptimizationbook.com/book/8.3-dangers-backtesting.html 54. Deflated Sharpe Ratio: Selection-Adjusted Evidence - Pineify, https://pineify.app/backtesting/deflated-sharpe-ratio 55. software - Marcos M. Lopez de Prado, https://www.quantresearch.org/Software.htm